Executive Summary — Surrogate Model Validation Report

Use case: UCFATIGUE_1 | 07 September 2026 | 2 model(s)

REQUIREMENTS PARTIALLY MET

Recommended model: GradientBoosting

Avg R^2: 0.9643 — Excellent

Q90 Accuracy — Partially met

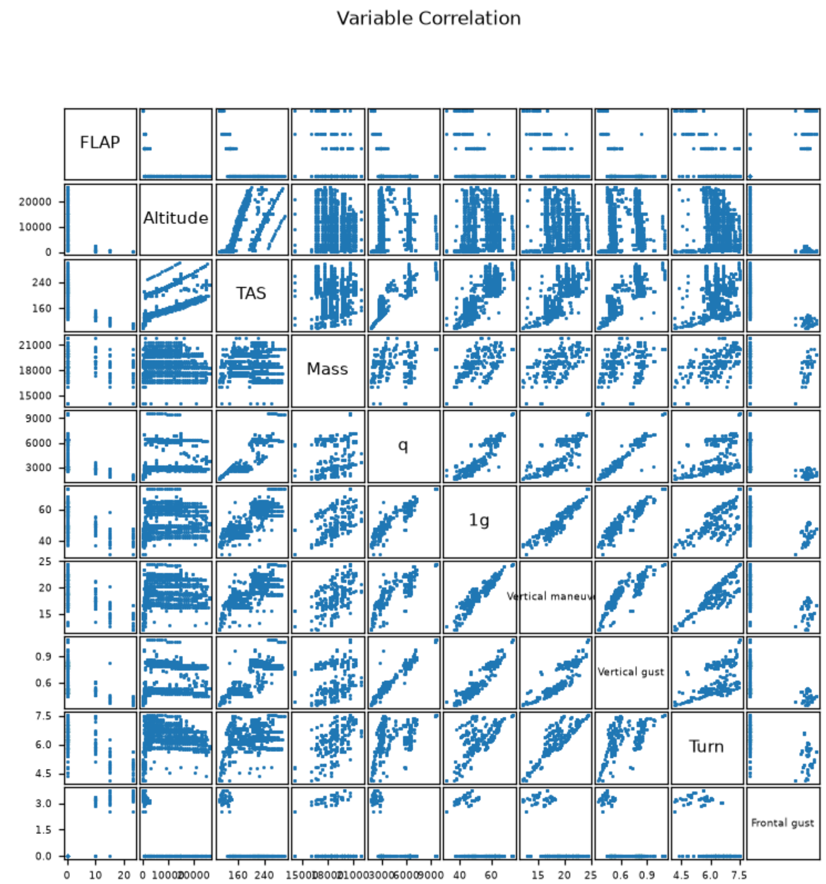
Pass: 1g, Vertical maneuver, Vertical gust, Turn, n0

Fail: Frontal gust, Giro

Project Overview

Use case	UCFATIGUE_1
Inputs	FLAP, Altitude, TAS, Mass, q, gamma, Type_segment, Xcg(%CMA)
Outputs	7: 1g, Vertical maneuver, Vertical gust, Turn, Frontal gust, n0, Giro
Train / Val / Test	80% / 10% / 10%
Accuracy target	Q90 < 10% relative error per output

Variable Correlation — Input vs Output



Data Statistics

Per-variable statistics over all data (train + val + test), as `Train_set.describe()` reports them in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum.

Inputs

	count	mean	std	min	P25	P50	P75	max
FLAP	869	2.2186	5.3696	0	0	0	0	23
Altitude	869	9294.6341	7701.9019	18	2,000	8,000	15,000	26,000
TAS	869	190.871	48.8676	101	144	197.5712	230	302
Mass	869	18874.7779	1324.6887	14,125	17,761	18,905	19,991	21,886
q	869	4510.2453	1845.3166	1652.3866	2881.2026	3579.9882	6375.4631	9637.8604
gamma	869	-0.2667	4.5734	-39.4508	-4.7042	0	2.9442	18.2194
Xcg(%CMA)	869	19.4107	4.0946	14.6	16.43	18.5	20.58	29.3

Input variables — Train_set.describe()

Outputs

	count	mean	std	min	P25	P50	P75	max
1g	869	53.6228	8.2962	31.561	47.364	51.778	61.558	73.308
Vertical maneuver	869	19.1377	2.4542	12.031	17.185	18.893	21.394	24.471
Vertical gust	869	0.6494	0.1681	0.3466	0.5097	0.575	0.806	1.092
Turn	869	6.3621	0.6415	4.122	6.01	6.382	6.83	7.53
Frontal gust	869	0.4806	1.1541	0	0	0	0	3.722
n0	869	1.9391	0.0777	1.73	1.9	1.94	1.98	2.19
Giro	869	1.9959	1.5329	0.662	1.299	1.587	1.874	7.599

Output variables — Train_set.describe()

Model Selection

Model	Avg R^2	Q90 pass (7)	KS pass (7)
MLP	0.5237	5/7	7/7
GradientBoosting ⋆	0.9643	5/7	2/7

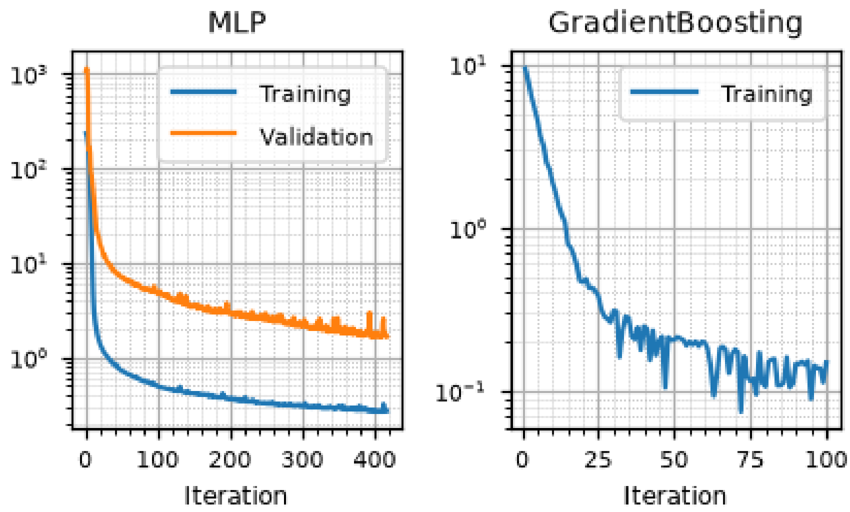
⋆ Recommended model. Q90 target: $< 10\%$. KS: $p \geq 0.05$ = no overfitting.

Accuracy Assessment (Q90 & R^2)

Output	MLP			GradientBoosting ⋆		
	R^2	Q90	Gap	R^2	Q90	Gap
1g	0.9160	0.0473	-0.0527	0.9797	0.0257	-0.0743
Vertical maneuver	0.9116	0.0554	-0.0446	0.9723	0.0318	-0.0682
Vertical gust	0.9377	0.0918	-0.0082	0.9914	0.0262	-0.0738
Turn	0.7544	0.0783	-0.0217	0.9286	0.0351	-0.0649
Frontal gust	0.9794	11.3300	+11.2300	0.9964	0.2795	+0.1795
n0	-0.7292	0.0723	-0.0277	0.9317	0.0149	-0.0851
Giro	-0.1042	0.9005	+0.8005	0.9499	0.2215	+0.1215

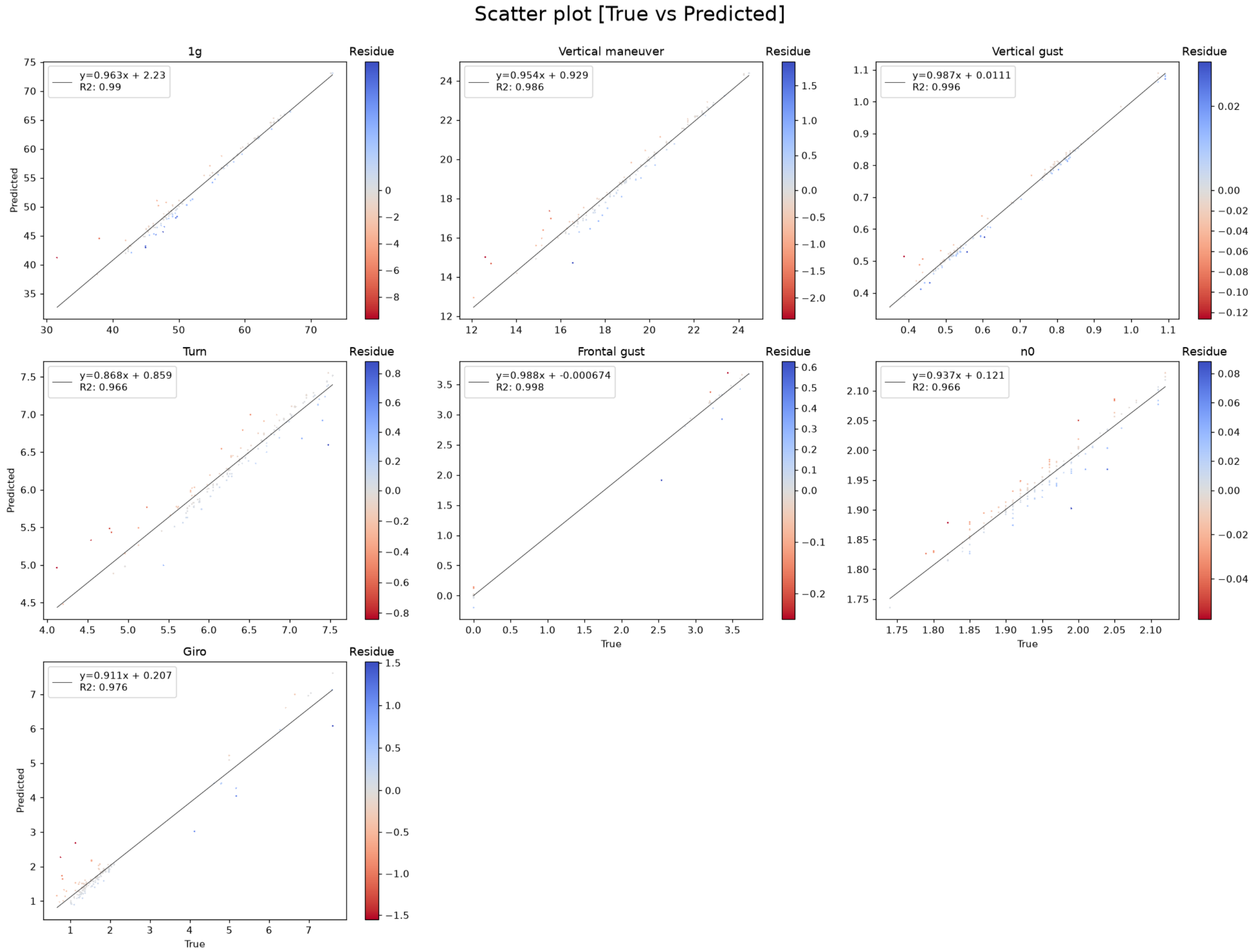
$R^2 \geq 0.95$ $R^2 \in [0.80, 0.95)$ $R^2 < 0.80$ or Q90 fail pass Gap=Q90−target ($< 0 \Rightarrow$ margin)

Training History



Training and validation loss per iteration, log scale. For models trained with early stopping, the validation loss is approximated from the recorded validation R^2 via $(1 - R^2) \cdot \mathrm{Var}(y)/2$. Gradient boosting reports per-stage deviance averaged over its per-output estimators.

Predicted vs True — GradientBoosting



Data Split Quality

Metric	Value	Pass
Residual voxel proportion ($\leq$0.05)	0.000	✓
Valid test proportion	0.931	
p-hacking test proportion ($\leq$0.05 warn)	0.006	✓
Isolated test proportion ($\leq$0.05 warn)	0.063	×
Chi^2 p-value ($\geq$0.05)	0.7969	✓

No p-hacking concern: only 0.6% of test points lie closer to a training point than training points are to each other, so the test error is not flattered. Isolated points: 6.3% of test points have no training point nearby, so part of the test error measures extrapolation rather than interpolation — the model may look worse there than it is in the region it was trained on.

Overfitting Check (KS Test) — GradientBoosting

Output	KS p-value
1g	0.004
Vertical maneuver	0.161
Vertical gust	0.042
Turn	0.044
Frontal gust	0.974
n0	0.032
Giro	0.017

$p \geq 0.05$ — no overfitting $p < 0.05$ — overfitting detected

Deep Analysis

UCFATIGUE_1

Winner Model — Full Validation

Deep Analysis — GradientBoosting

Mirrors `validation_template.ipynb` — all computations from validation CSVs

Data Overview

Statistical summary of inputs and outputs over all data (train + val + test), as reported by `describe()` in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum per variable. All figures in this part are drawn by `validationlib`, matching the extended validation notebook.

Input Statistics

	count	mean	std	min	P25	P50	P75	max
FLAP	869	2.2186	5.3696	0	0	0	0	23
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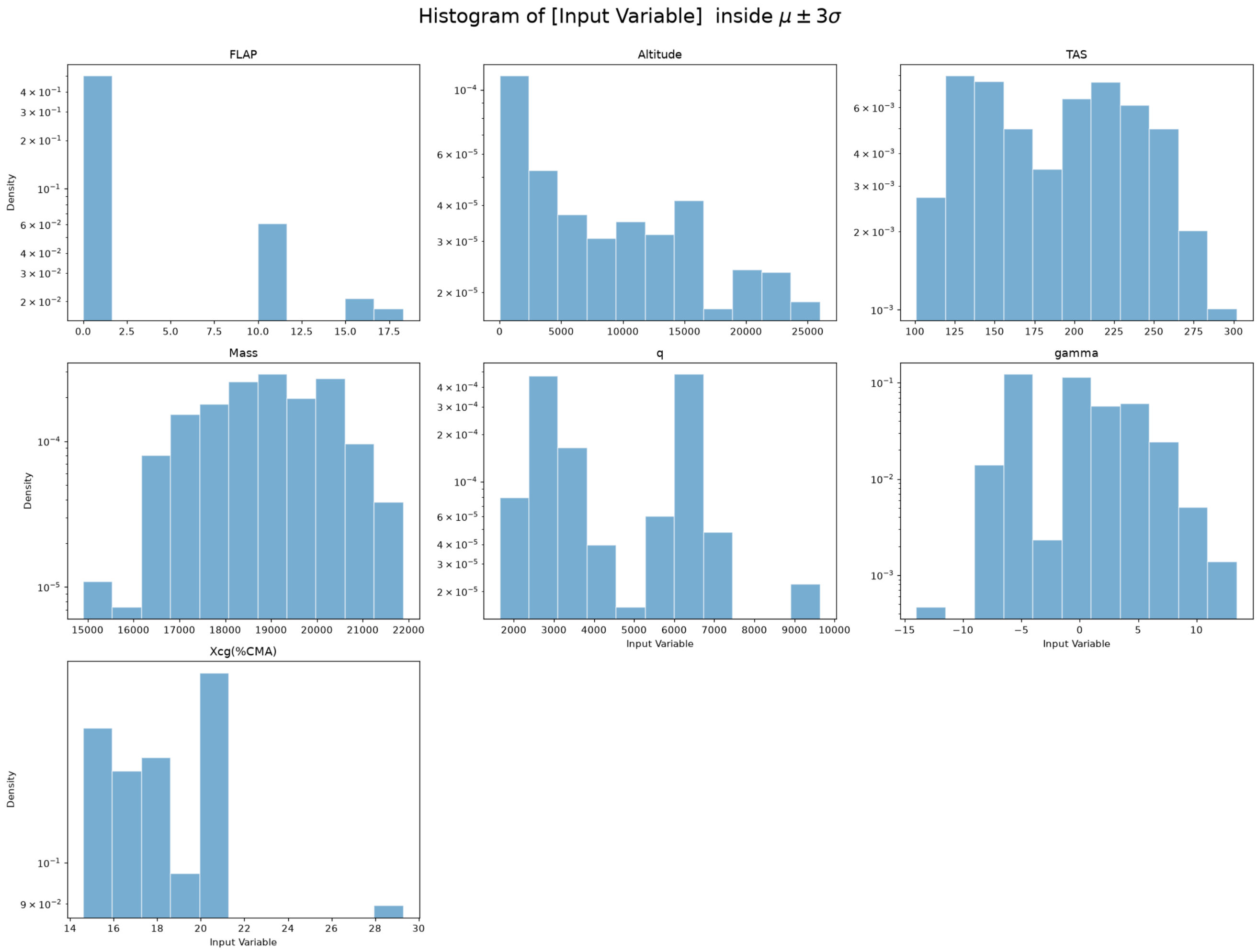
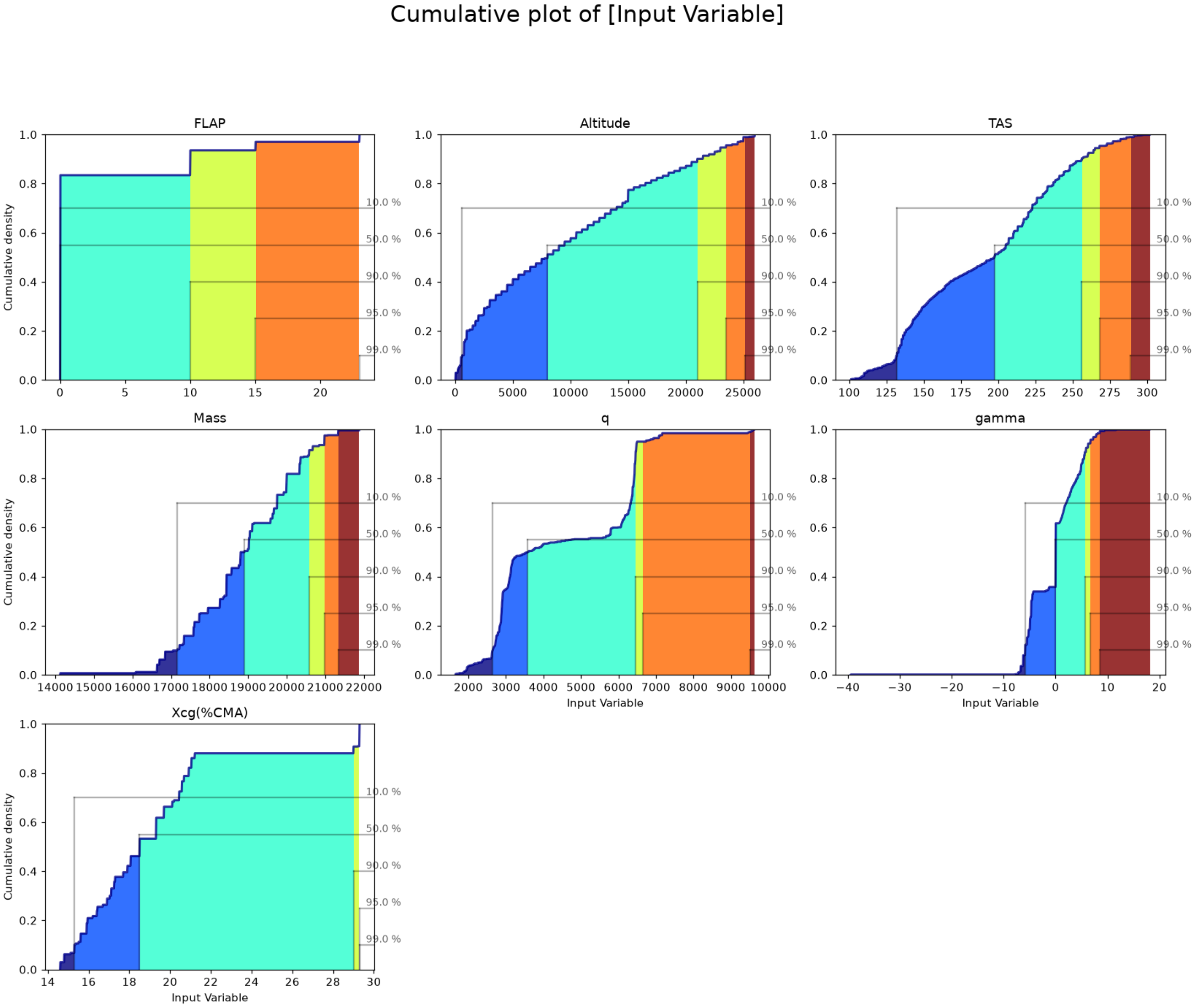
Xcg(%CMA)	869	19.4107	4.0946	14.6	16.43	18.5	20.58	29.3

Input variables — Train_set.describe()

Input Histograms



Input Cumulative Distributions



Output Statistics

	count	mean	std	min	P25	P50	P75	max
1g	869	53.6228	8.2962	31.561	47.364	51.778	61.558	73.308
Vertical maneuver	869	19.1377	2.4542	12.031	17.185	18.893	21.394	24.471
Vertical gust	869	0.6494	0.1681	0.3466	0.5097	0.575	0.806	1.092
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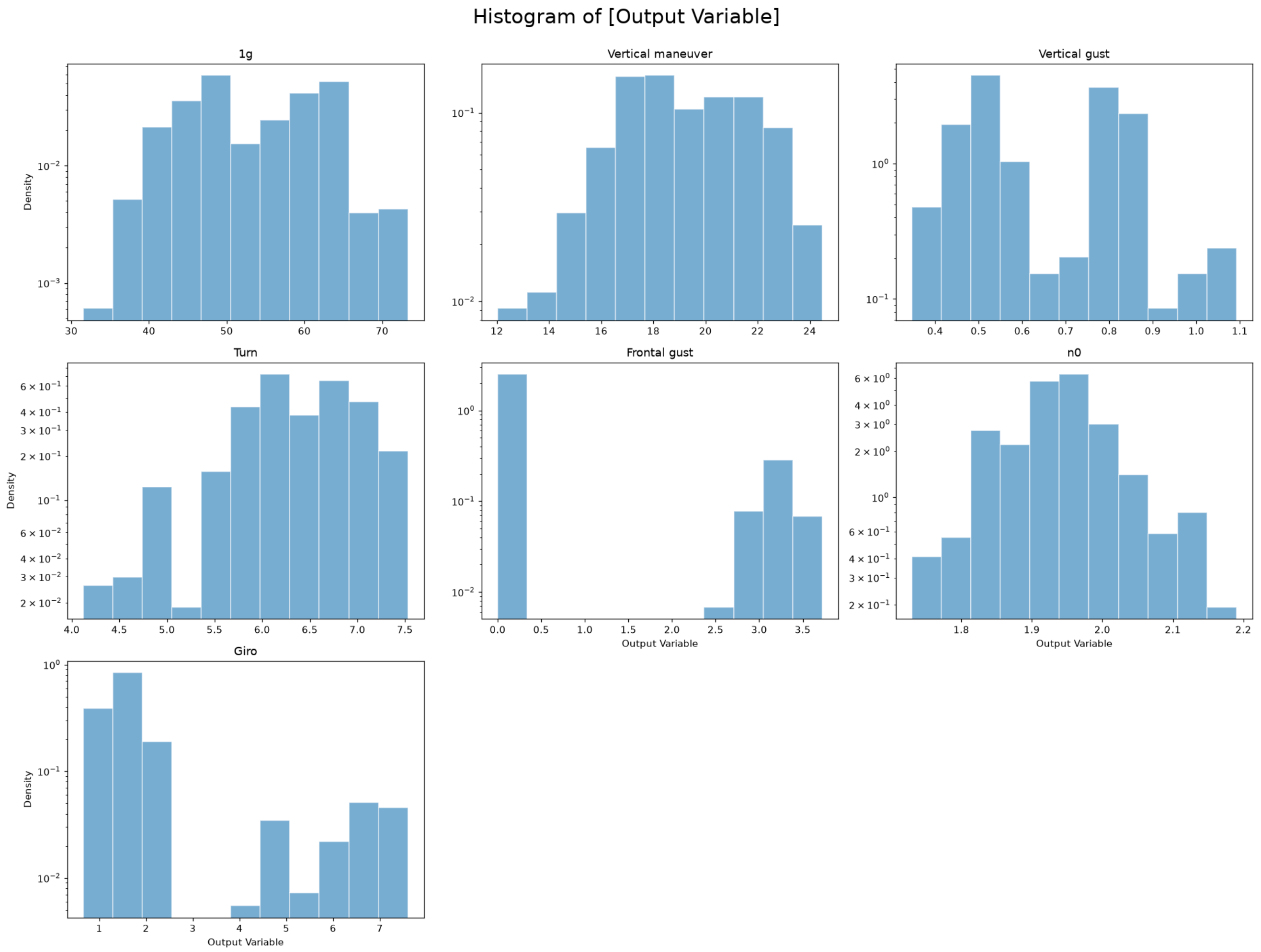
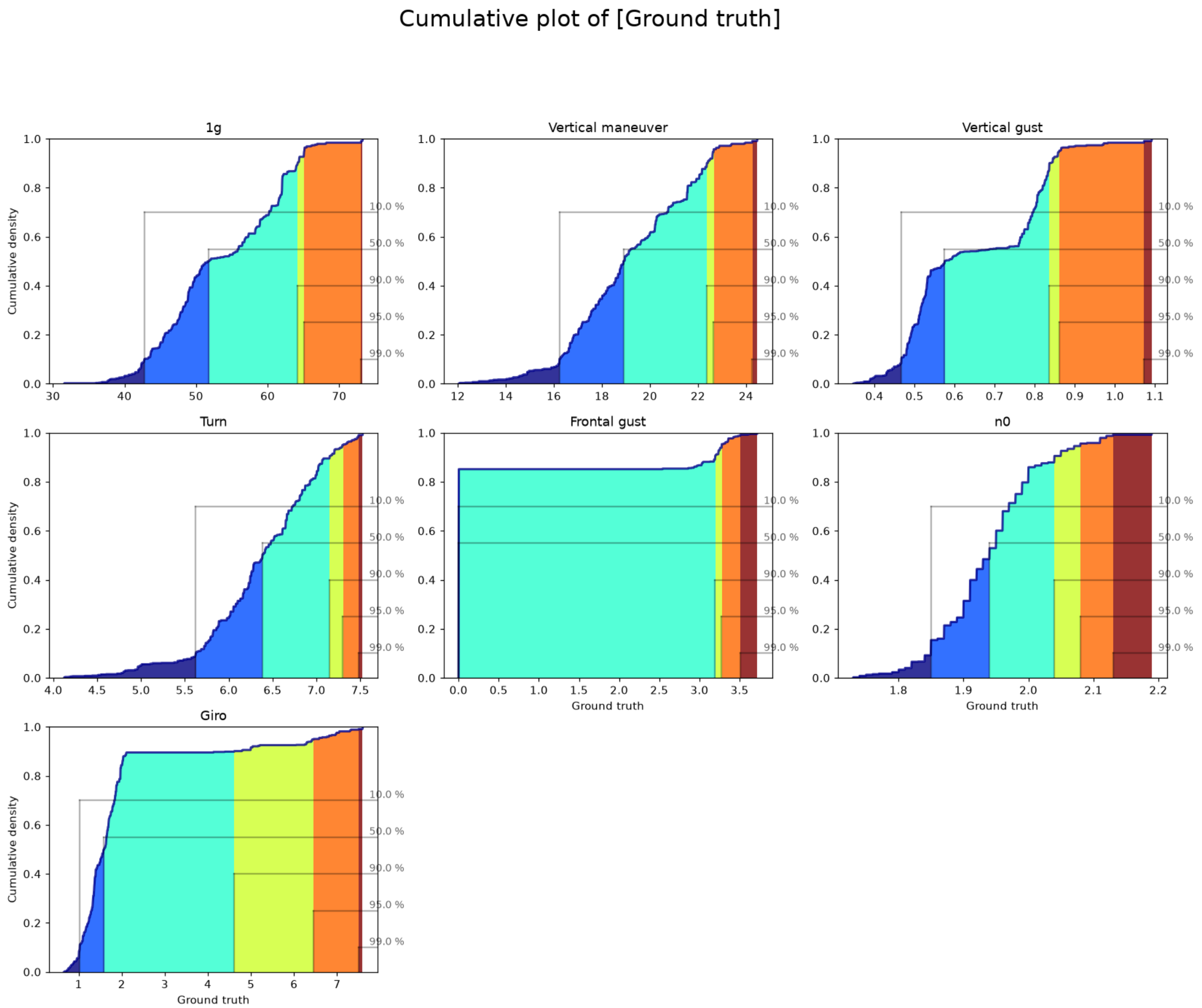
Output variables — Train_set.describe()

Output Histograms



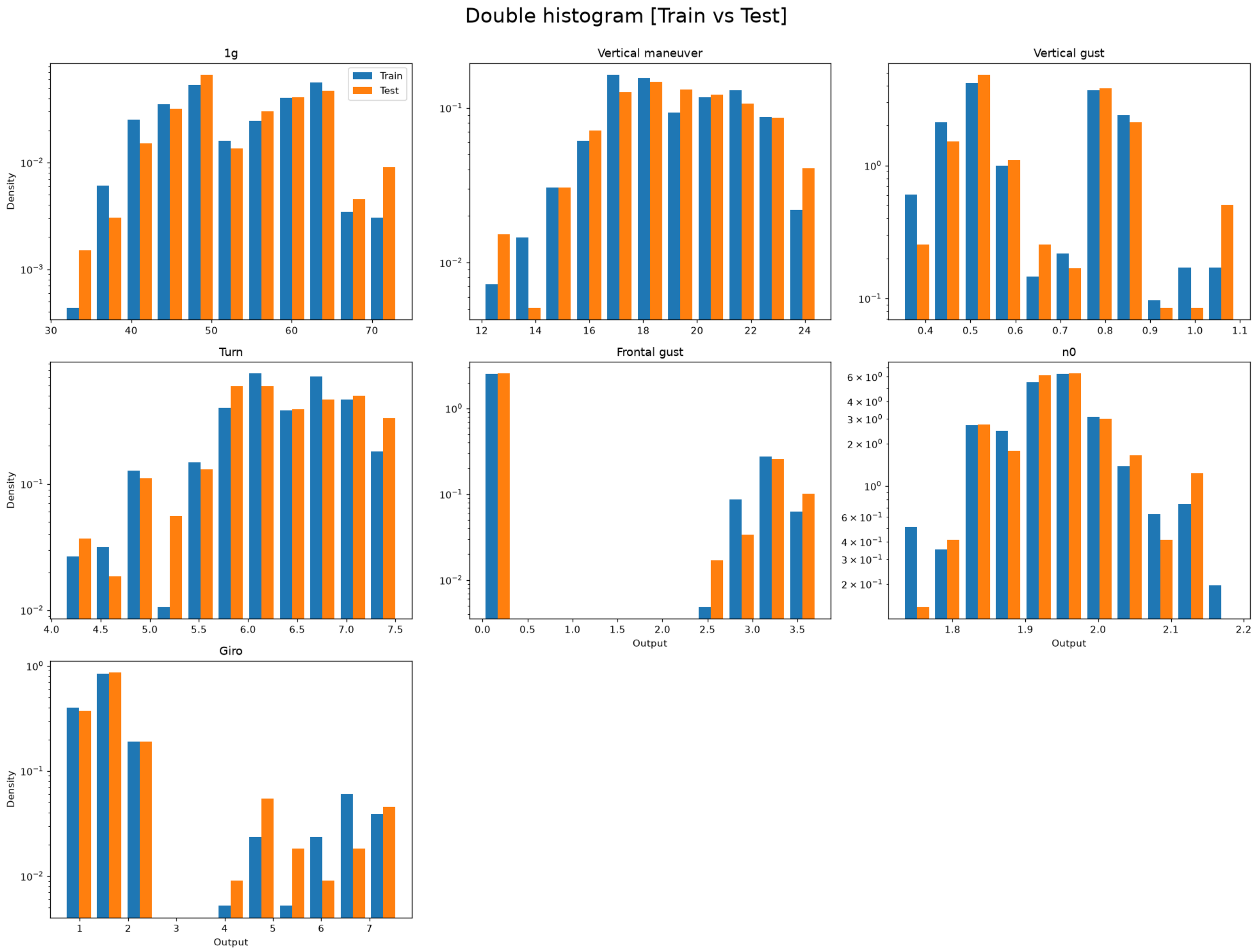
Output Cumulative Distributions



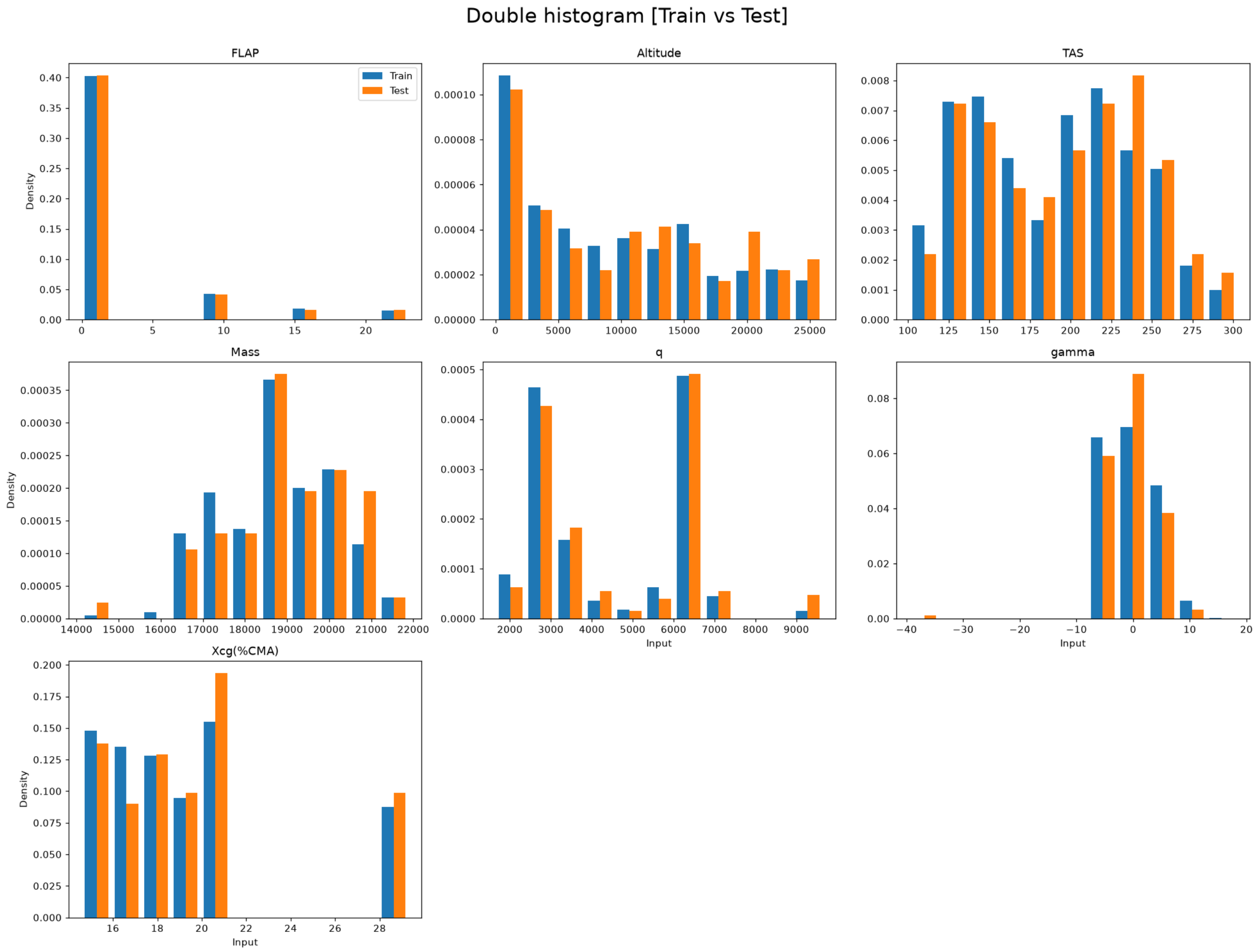
Train-Test Split Analysis

A well-designed split produces training and test distributions that are similar but not identical. The Kolmogorov-Smirnov (KS) test and the Anderson-Darling (AD) test are used to assess distributional similarity. H_0: both sets come from the same distribution. $p < 0.05$ rejects H_0 (undesirable if too low; undesirable if the split was done by stratification and p is too high).

Output Distributions: Train vs Test



Input Distributions: Train vs Test



KS and AD Test Results

	KS	AD
1g	0.126	0.1656
Vertical maneuver	0.9577	0.25
Vertical gust	0.4768	0.25
Turn	0.4441	0.1512
Frontal gust	1	0.25
n0	0.7097	0.25
Giro	0.4079	0.25

Train vs test p-values. Red: $p < 0.05$ (distributions differ).

Split Quality — Voxel Tesselation Proximity (p-hacking check)

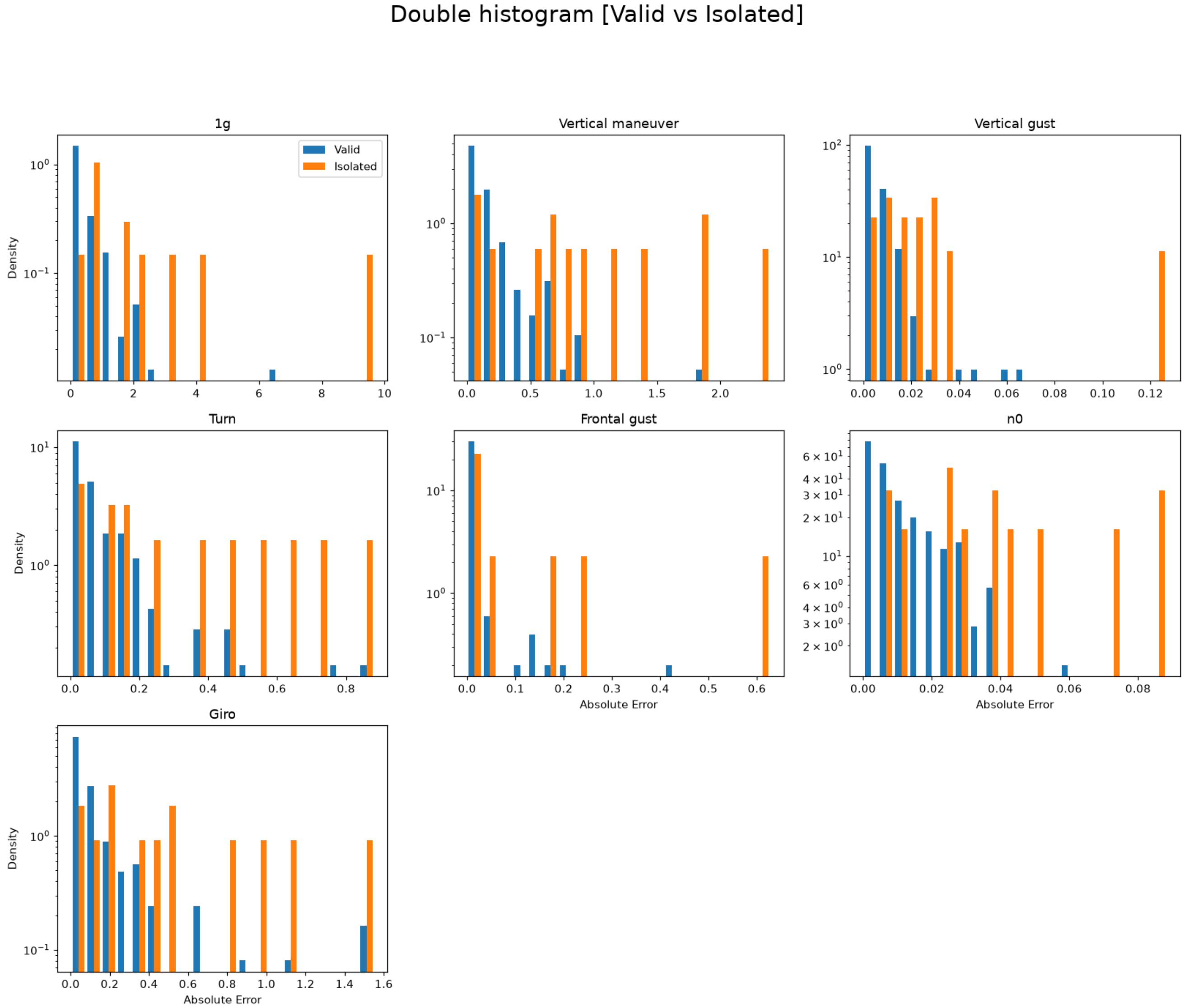
Beyond matching distributions, a split can still flatter the model if test points sit right next to training points (p-hacking) or probe regions the model never saw (isolated, residual voxel). The VTPM method from `validationlib` (SF_9 cell 9.0) classifies every test point.

Metric	Value	Pass
Residual voxel proportion ($\leq$0.05)	0.000	✓
Valid test proportion	0.931	
p-hacking test proportion ($\leq$0.05 warn)	0.006	✓
Isolated test proportion ($\leq$0.05 warn)	0.063	×
Chi^2 p-value ($\geq$0.05)	0.7969	✓

No p-hacking concern: only 0.6% of test points lie closer to a training point than training points are to each other, so the test error is not flattered. Isolated points: 6.3% of test points have no training point nearby, so part of the test error measures extrapolation rather than interpolation — the model may look worse there than it is in the region it was trained on.

Error Distribution: Valid vs Isolated Test Points

Same comparison for the test points with no training neighbour.



	AD	KS	KW
1g	0.001	3.19e-07	3.36e-06
Vertical maneuver	0.001	1e-05	9.16e-05
Vertical gust	0.001	0.000205	9.84e-06
Turn	0.001	0.0015	0.000255
Frontal gust	0.0016	0.0605	0.0069
n0	0.001	9.08e-06	9.59e-06
Giro	0.001	8.38e-05	0.000123

Valid ($n = 160$) vs Isolated ($n = 14$) absolute-error distributions, on a subsample of the test set. Red: $p < 0.05$ (the flagged points behave differently).

Error Quantification — P(E)

Two error metrics are studied:

- Residue: $e_i = y_i - \hat{y}_i$. A centred ($\mu \approx 0$), symmetric distribution indicates an unbiased model.
- Absolute Error: $|e_i|$. Always non-negative; its Q90 must satisfy the accuracy requirement.

Residue Statistics Table

Bootstrap confidence bounds shown as superscript (upper) and subscript (lower).

	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
1g	174	-9.66	1.97	-0.21	$^{-0.0724}_{-0.378}$	-0.0456	$^{0.0019}_{-0.127}$	1.16	$^{1.52}_{0.679}$	0.594	$^{0.712}_{0.408}$	29.2	$^{46.8}_{5.69}$	-4.36	$^{-0.934}_{-5.52}$
Vertical maneuver	174	-2.39	1.85	-0.0341	$^{0.0244}_{-0.079}$	-0.0074	$^{0.0265}_{-0.0332}$	0.419	$^{0.496}_{0.312}$	0.224	$^{0.263}_{0.193}$	10.8	$^{15.8}_{6.01}$	-1.61	$^{0.541}_{-2.96}$
Vertical gust	174	-0.127	0.0307	-0.0024	$^{-0.000455}_{-0.0046}$	-0.0013	$^{0.000254}_{-0.0025}$	0.0157	$^{0.0209}_{0.0114}$	0.0097	$^{0.0127}_{0.0085}$	23.7	$^{35.6}_{4.22}$	-3.64	$^{-1.14}_{-4.55}$
Turn	174	-0.84	0.881	-0.0176	$^{0.0077}_{-0.0391}$	-0.0031	$^{0.0136}_{-0.0131}$	0.184	$^{0.216}_{0.142}$	0.0983	$^{0.128}_{0.0713}$	8	$^{12.3}_{4.72}$	-0.788	$^{1.02}_{-2.56}$
Frontal gust	174	-0.249	0.628	0.0061	$^{0.0153}_{-0.0023}$	9.16e-05	$^{9.16e-05}_{9.16e-05}$	0.068	$^{0.0992}_{0.0325}$	1.39e-17	$^{0.00033}_{0}$	48.3	$^{96.2}_{17.4}$	5.57	$^{8.36}_{-2.71}$
n0	174	-0.0583	0.0883	0.0022	$^{0.0048}_{-0.000258}$	0.0014	$^{0.0031}_{-0.002}$	0.0192	$^{0.023}_{0.0155}$	0.015	$^{0.0202}_{0.013}$	4.62	$^{6.66}_{1.86}$	0.979	$^{1.61}_{-0.107}$
Giro	174	-1.55	1.51	-0.034	$^{-0.0016}_{-0.0769}$	-0.0113	$^{-0.0019}_{-0.0271}$	0.308	$^{0.394}_{0.209}$	0.142	$^{0.176}_{0.0961}$	11.1	$^{17.9}_{6.48}$	-0.253	$^{1.65}_{-2.37}$

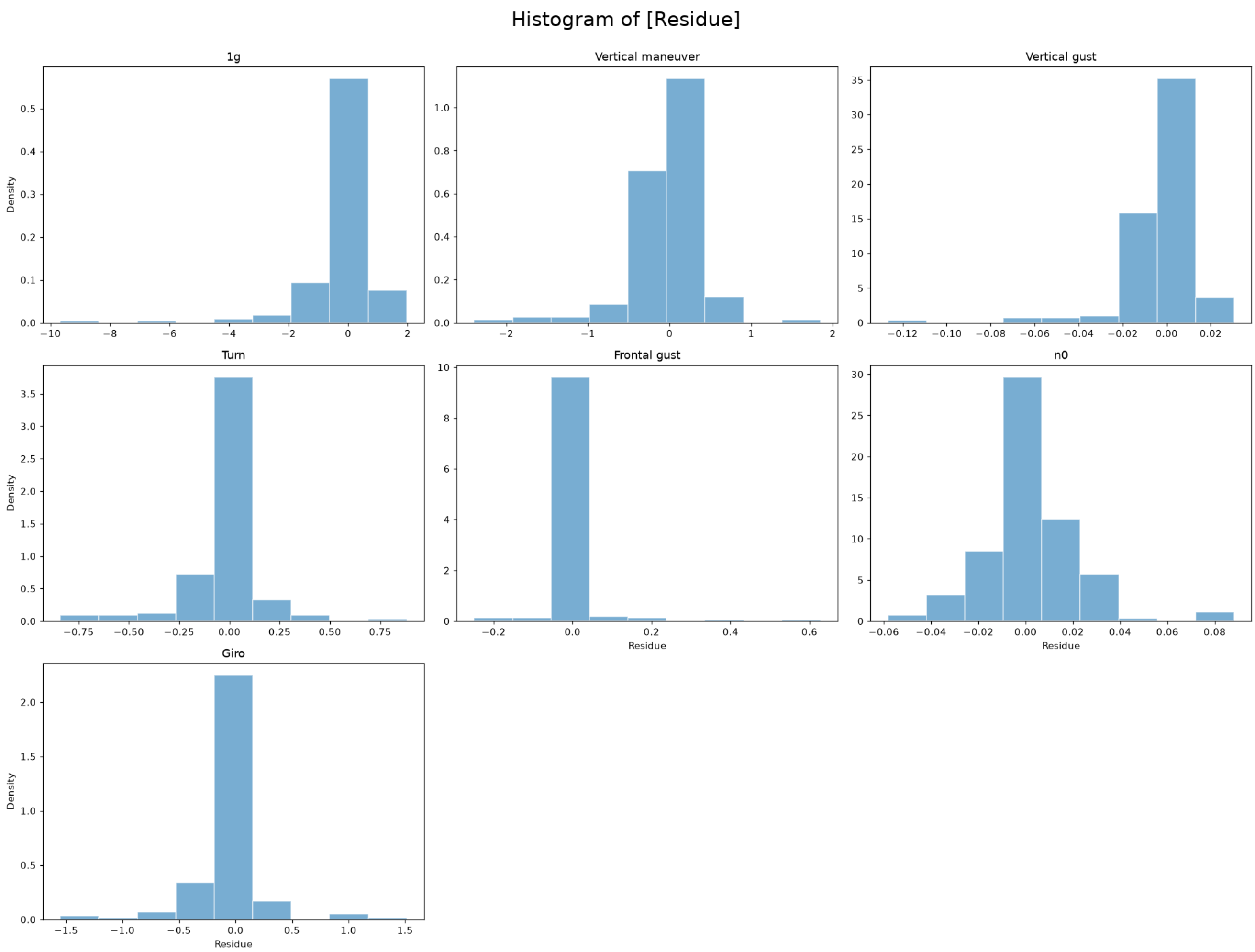
Residue statistics with bootstrap confidence intervals (superscript/subscript bounds).

Absolute Error Statistics Table

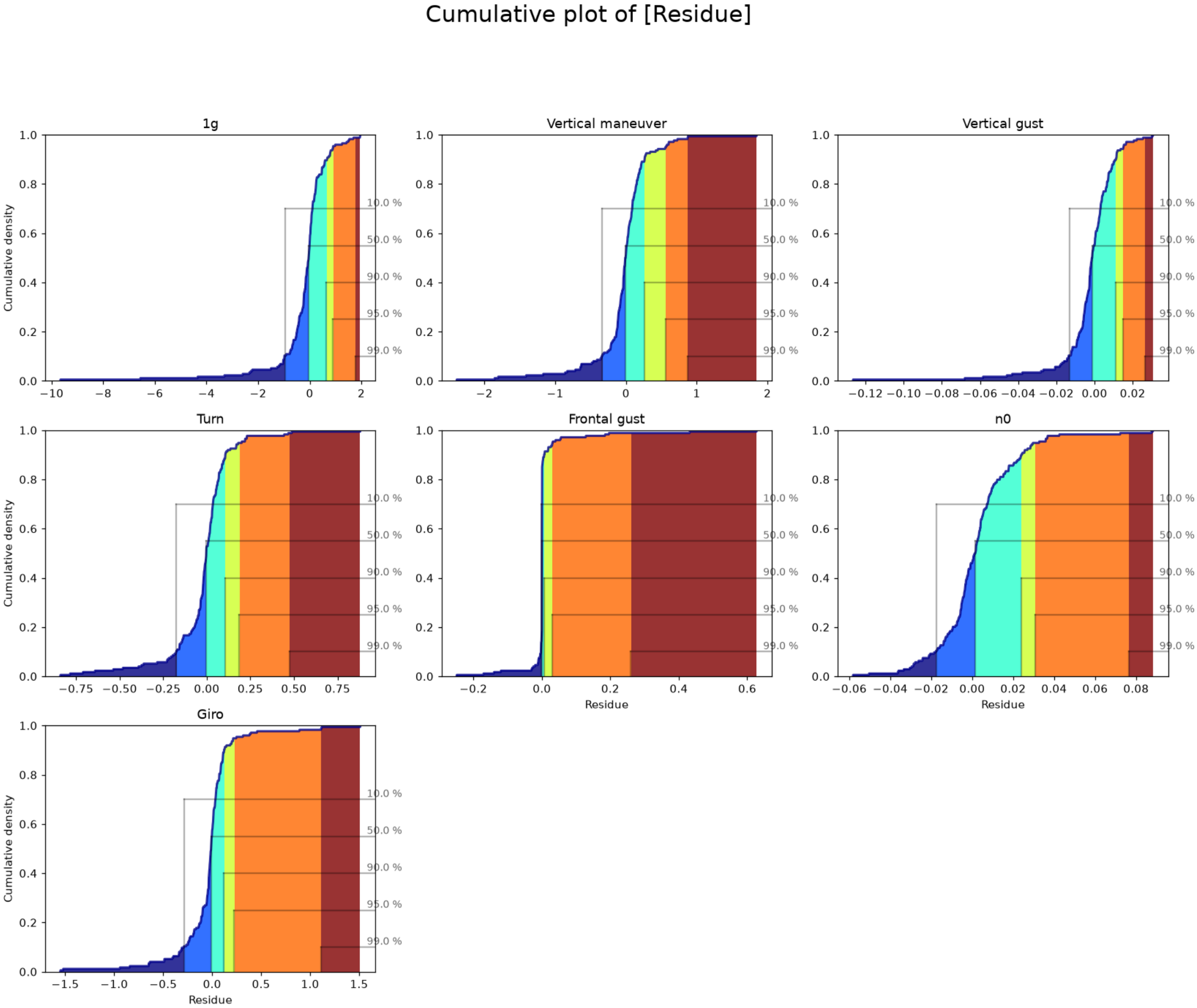
	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
1g	174	0.0028	9.66	0.572	$^{0.721}_{0.451}$	0.246	$^{0.363}_{0.223}$	1.03	$^{1.43}_{0.526}$	0.55	$^{0.68}_{0.412}$	39.7	$^{63.1}_{6.25}$	5.57	$^{6.94}_{2.21}$
Vertical maneuver	174	0.000887	2.39	0.229	$^{0.284}_{0.187}$	0.113	$^{0.137}_{0.0935}$	0.353	$^{0.45}_{0.245}$	0.177	$^{0.266}_{0.142}$	13.9	$^{22.9}_{7.02}$	3.45	$^{4.2}_{2.47}$
Vertical gust	174	3.12e-05	0.127	0.0088	$^{0.0109}_{0.0072}$	0.0055	$^{0.0061}_{0.0045}$	0.0132	$^{0.0192}_{0.0075}$	0.0073	$^{0.0093}_{0.0064}$	38.1	$^{53.5}_{7.19}$	5.28	$^{6.36}_{2.44}$
Turn	174	0.0016	0.881	0.105	$^{0.126}_{0.0851}$	0.0513	$^{0.0633}_{0.0376}$	0.153	$^{0.184}_{0.111}$	0.0955	$^{0.134}_{0.0714}$	9.81	$^{14.5}_{5.55}$	3	$^{3.49}_{2.38}$
Frontal gust	174	9.16e-05	0.628	0.0162	$^{0.0265}_{0.0081}$	9.16e-05	$^{9.16e-05}_{9.16e-05}$	0.0663	$^{0.0968}_{0.0281}$	0.002	$^{0.0059}_{0.000381}$	49.2	$^{86}_{17.1}$	6.54	$^{8.64}_{4.16}$
n0	174	5.6e-05	0.0883	0.013	$^{0.0148}_{0.0111}$	0.0076	$^{0.0099}_{0.0058}$	0.0143	$^{0.0174}_{0.011}$	0.0141	$^{0.0178}_{0.0113}$	8.74	$^{11.9}_{2.44}$	2.52	$^{2.93}_{1.47}$
Giro	174	0.0012	1.55	0.158	$^{0.189}_{0.12}$	0.0569	$^{0.0859}_{0.0448}$	0.266	$^{0.332}_{0.193}$	0.149	$^{0.185}_{0.0903}$	12.4	$^{21.8}_{7.24}$	3.36	$^{4.25}_{2.73}$

Absolute-error statistics with bootstrap confidence intervals.

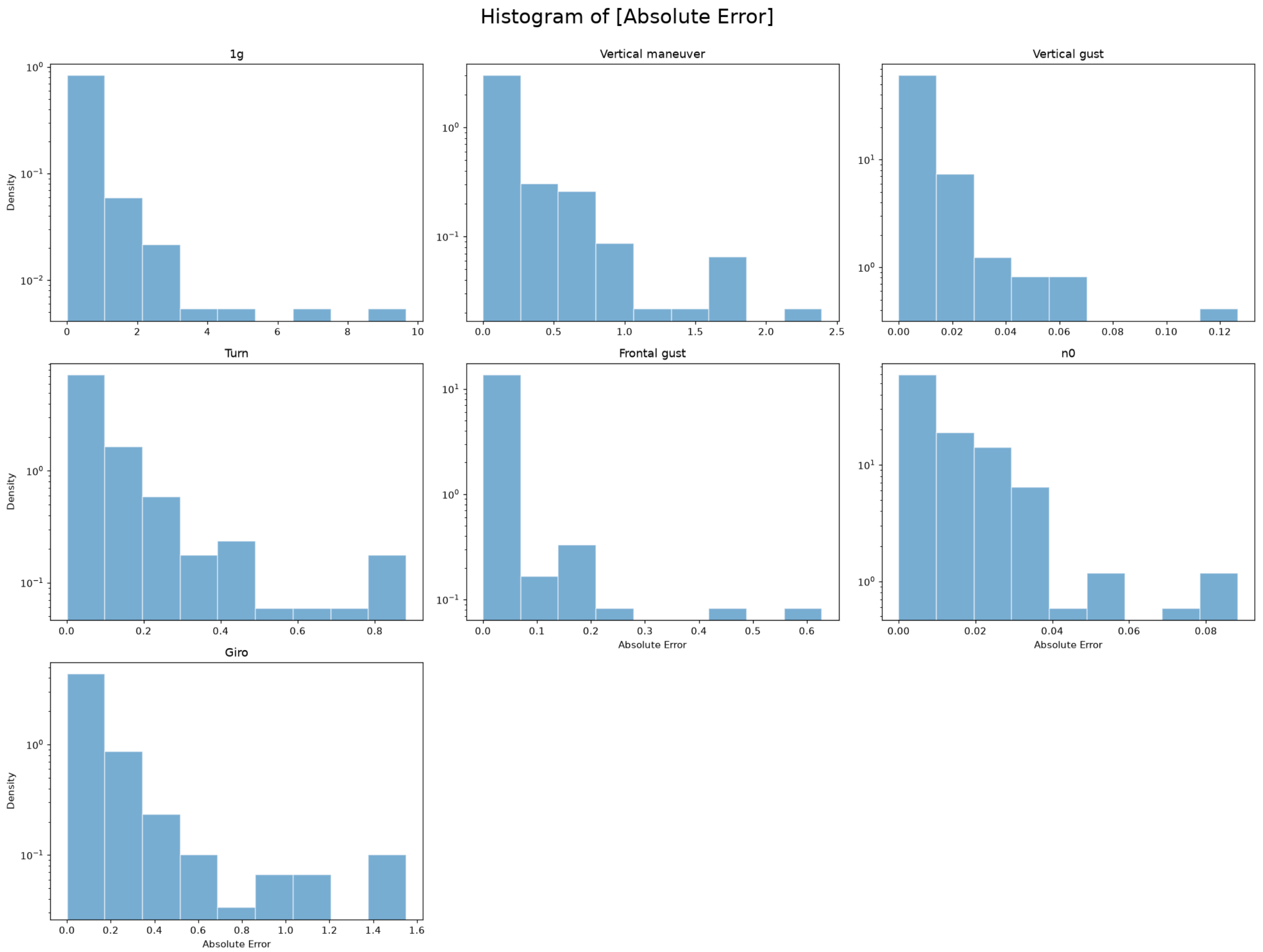
Residue Histograms



Residue CDF


Cumulative plot of [Residue]
1g
Vertical maneuver
Vertical gust
Turn
Frontal gust
n0
Giro
Cumulative density
Residue
10.0 %
50.0 %
90.0 %
95.0 %
99.0 %

Absolute Error Histograms



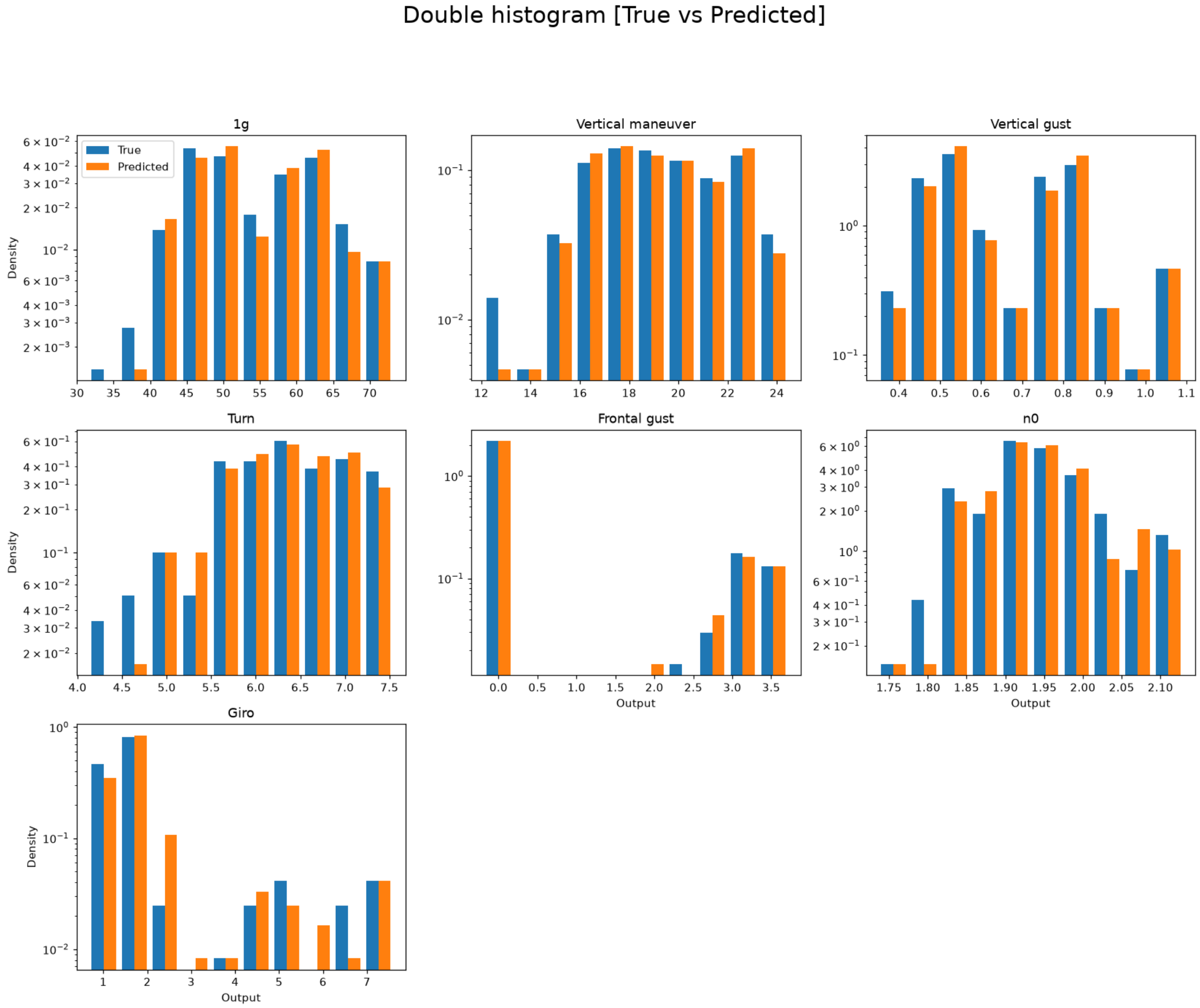
Q90 Accuracy Requirement

Output	Q90	Target	Status
1g	0.0257	10%	PASS
Vertical maneuver	0.0318	10%	PASS
Vertical gust	0.0267	10%	PASS
Turn	0.0351	10%	PASS
Frontal gust	0.0668	10%	PASS
n0	0.0150	10%	PASS
Giro	0.2236	10%	FAIL

$p \geq 0.05$ — pass $p < 0.05$ — fail

True vs Predicted Distribution Comparison

AD and KS tests comparing the distribution of true vs. predicted values on the test set. A significant difference indicates the model is not reproducing the output distribution faithfully.

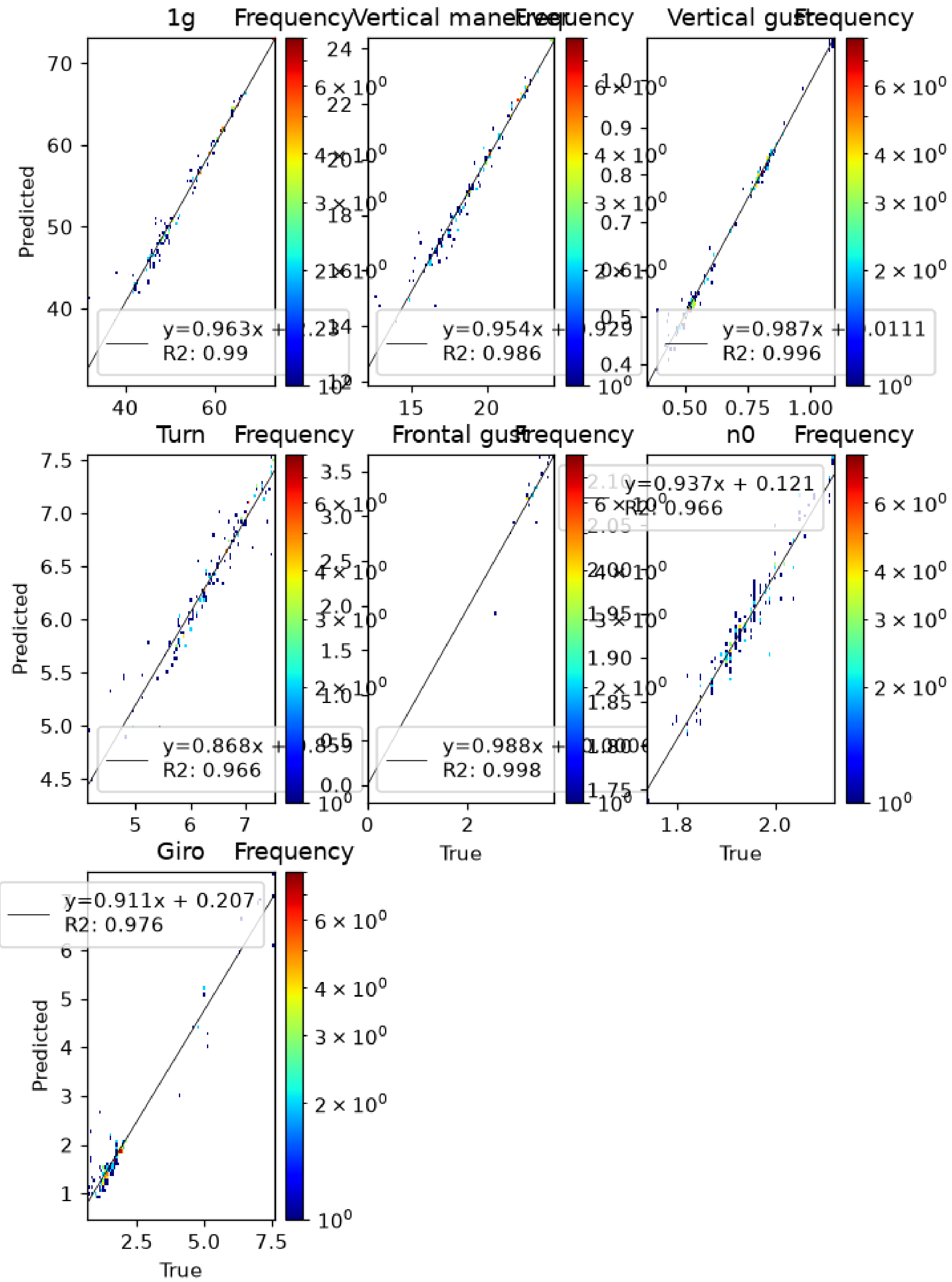


	AD	KS
1g	0.25	0.9932
Vertical maneuver	0.25	0.9372
Vertical gust	0.25	0.9748
Turn	0.25	0.9372
Frontal gust	0.001	5.39e-46
n0	0.25	0.6278
Giro	0.2143	0.1239

True vs predicted distribution tests. Red: $p < 0.05$.

True vs Predicted — 2D Histogram

Log-scaled frequency with a linear fit; the normalized slope should be close to 1.



P(E|X) — Error Conditional on Inputs

We study whether the model error depends on the input variables. Two approaches are used: (1) a Kruskal-Wallis (KW) test on binned inputs (Sturges rule for bin count), and (2) violin plots showing the error distribution within each input bin.

Significant input-conditional bias ($p < 0.05$) detected for: FLAP→Turn, Mass→1g, Mass→Frontal gust, Mass→n0, q→1g, q→Vertical maneuver, q→Vertical gust, q→Turn, q→Giro, gamma→Frontal gust.... These inputs explain residual variance — consider polynomial features or interaction terms.

Bias Detection Table (KW p-values)

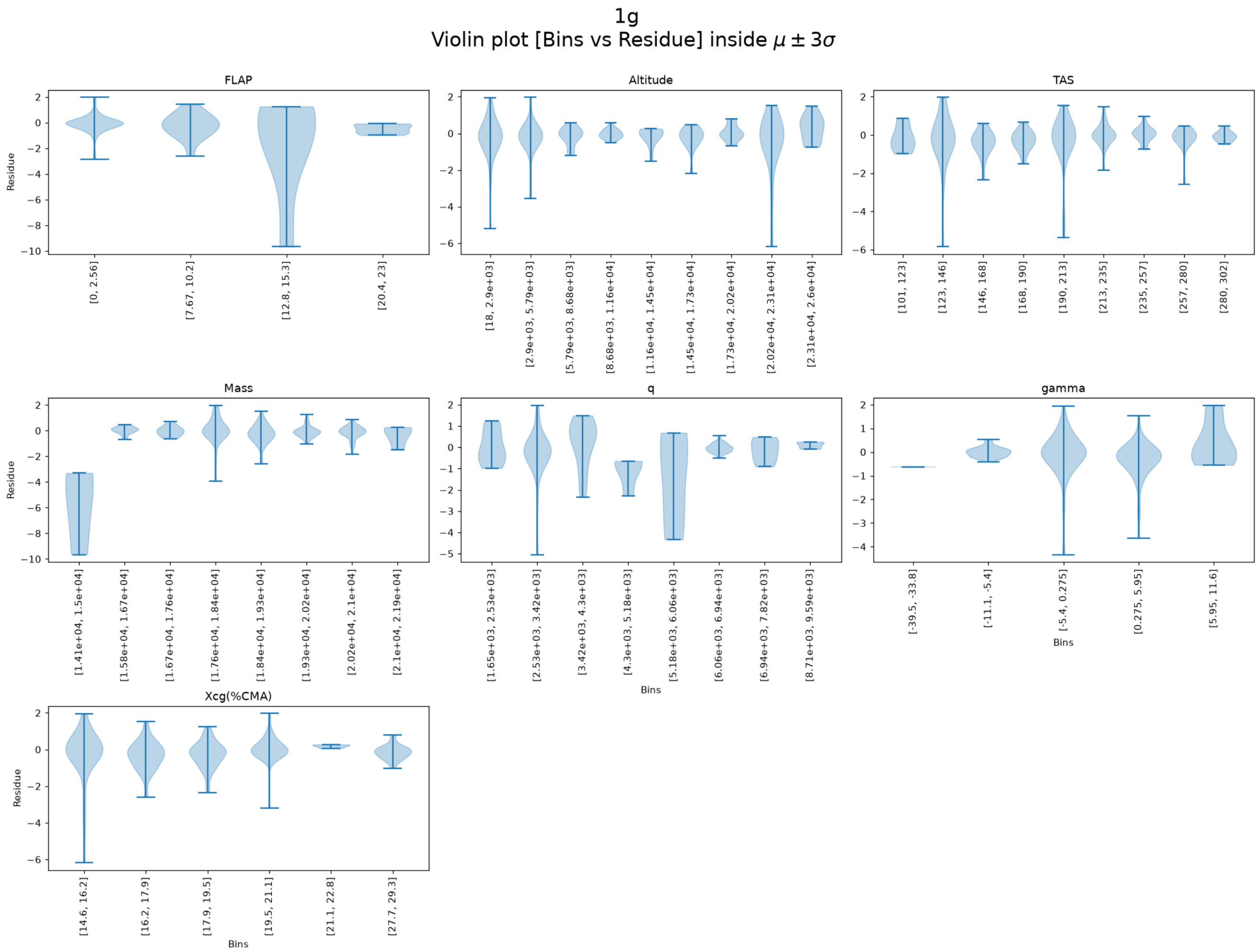
Green ($p \geq 0.05$) = residue uniform across input bins; red ($p < 0.05$) = significant bias.

	1g	Vertical maneuver	Vertical gust	Turn	Frontal gust	n0	Giro
FLAP_binned	0.2066	0.061	0.1567	0.0058	0.1015	0.7591	0.3727
Altitude_binned	0.7601	0.6686	0.2077	0.4912	0.9794	0.6757	0.1062
TAS_binned	0.3739	0.8346	0.7359	0.0718	0.5515	0.2738	0.3335
Mass_binned	0.0398	0.1396	0.781	0.0898	0.001	0.0157	0.1063
q_binned	0.0291	0.0041	0.0044	0.0469	0.1181	0.2334	4.54e-05
gamma_binned	0.0574	0.6324	0.7522	0.2503	0.0023	0.0561	0.2246
Xcg(%CMA)_binned	0.0479	0.4164	0.3997	0.3511	0.0087	0.5116	0.0382

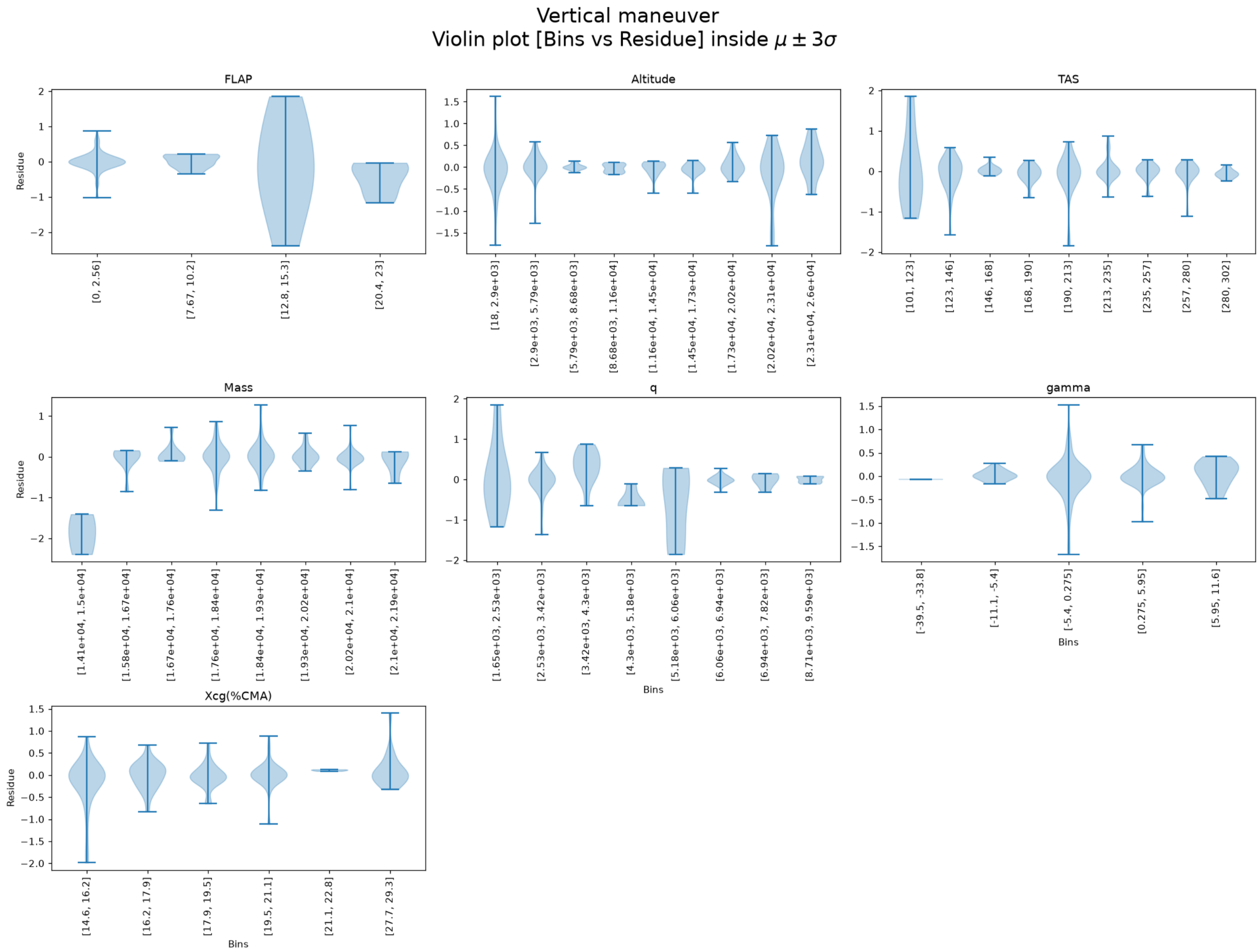
Kruskal-Wallis p-values on Sturges-binned inputs. Red: $p < 0.05$ (error depends on that input).

Residue Violin Plots per Output (binned by input)

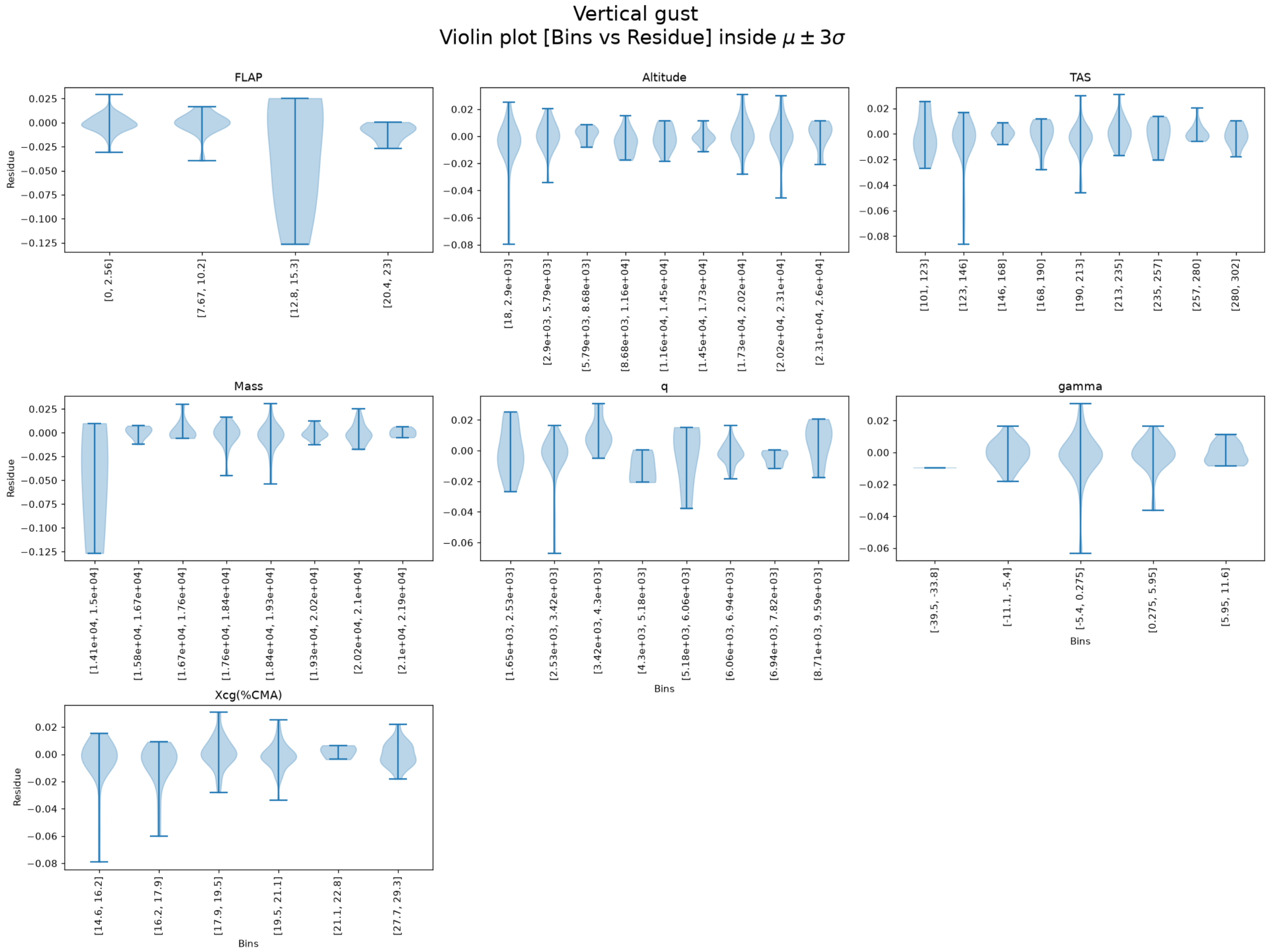
Each panel shows the residue distribution for each input bin. A flat median line (dashed at 0) and symmetric violins indicate absence of bias.
Residue — 1g



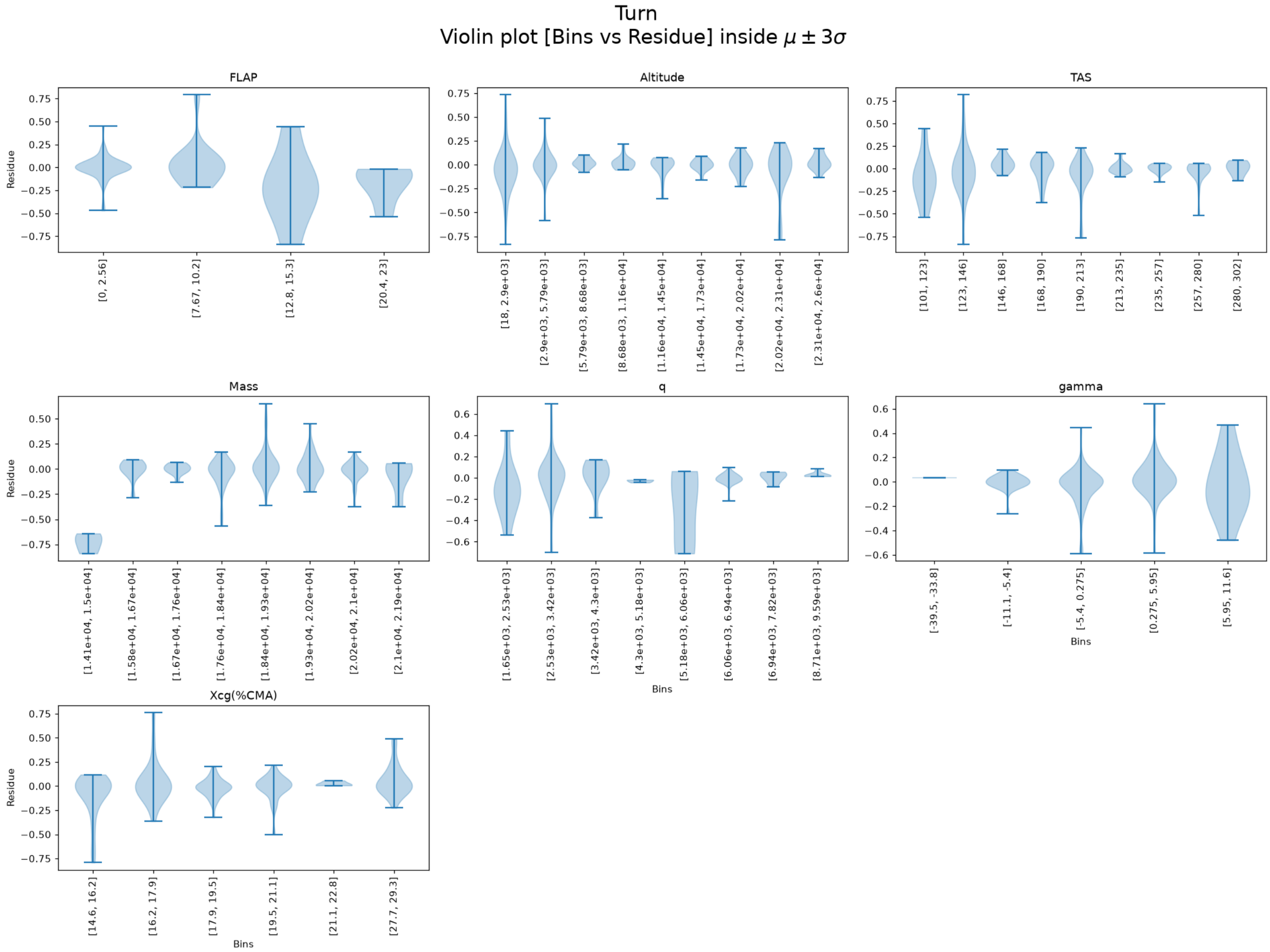
Residue — Vertical maneuver



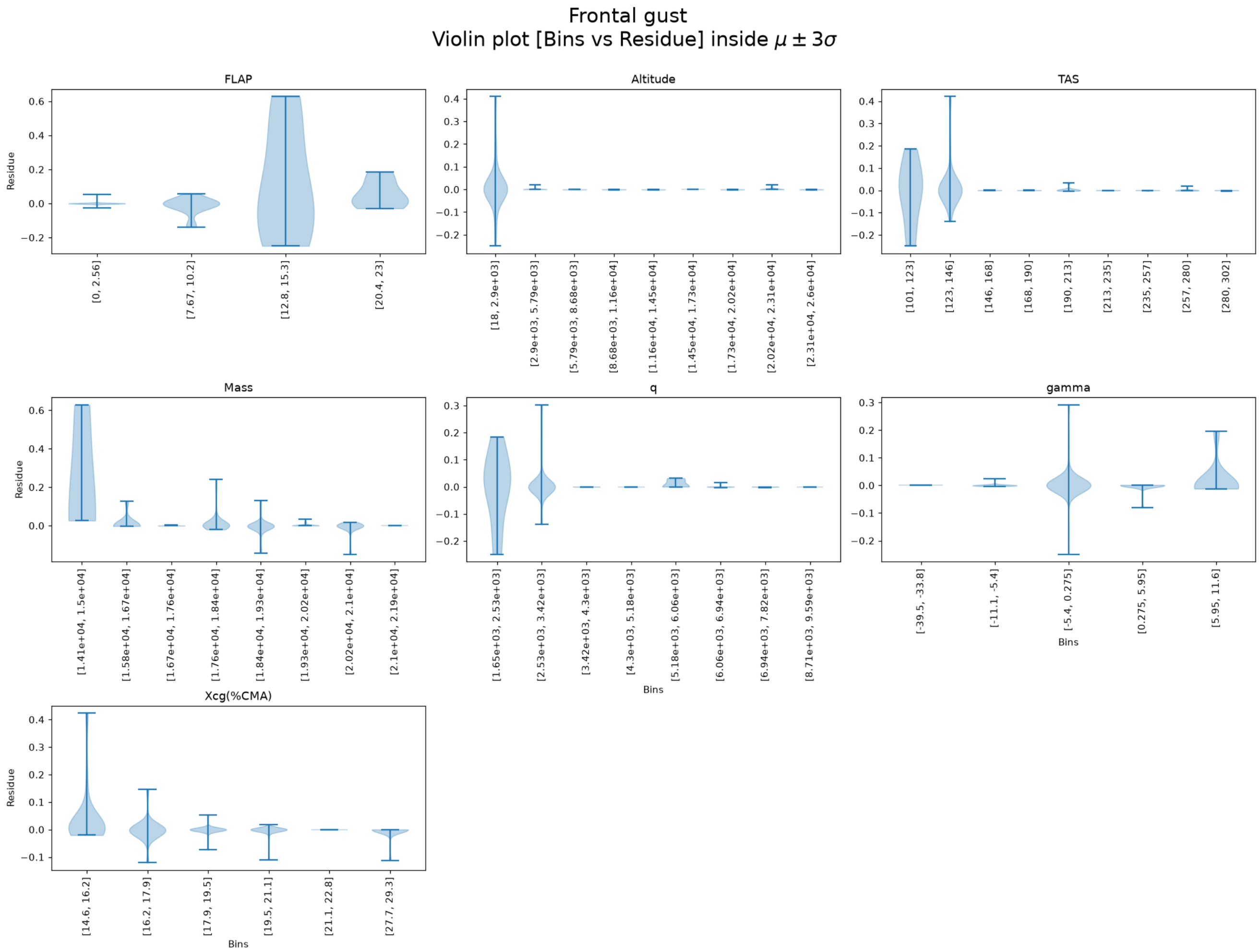
Residue — Vertical gust



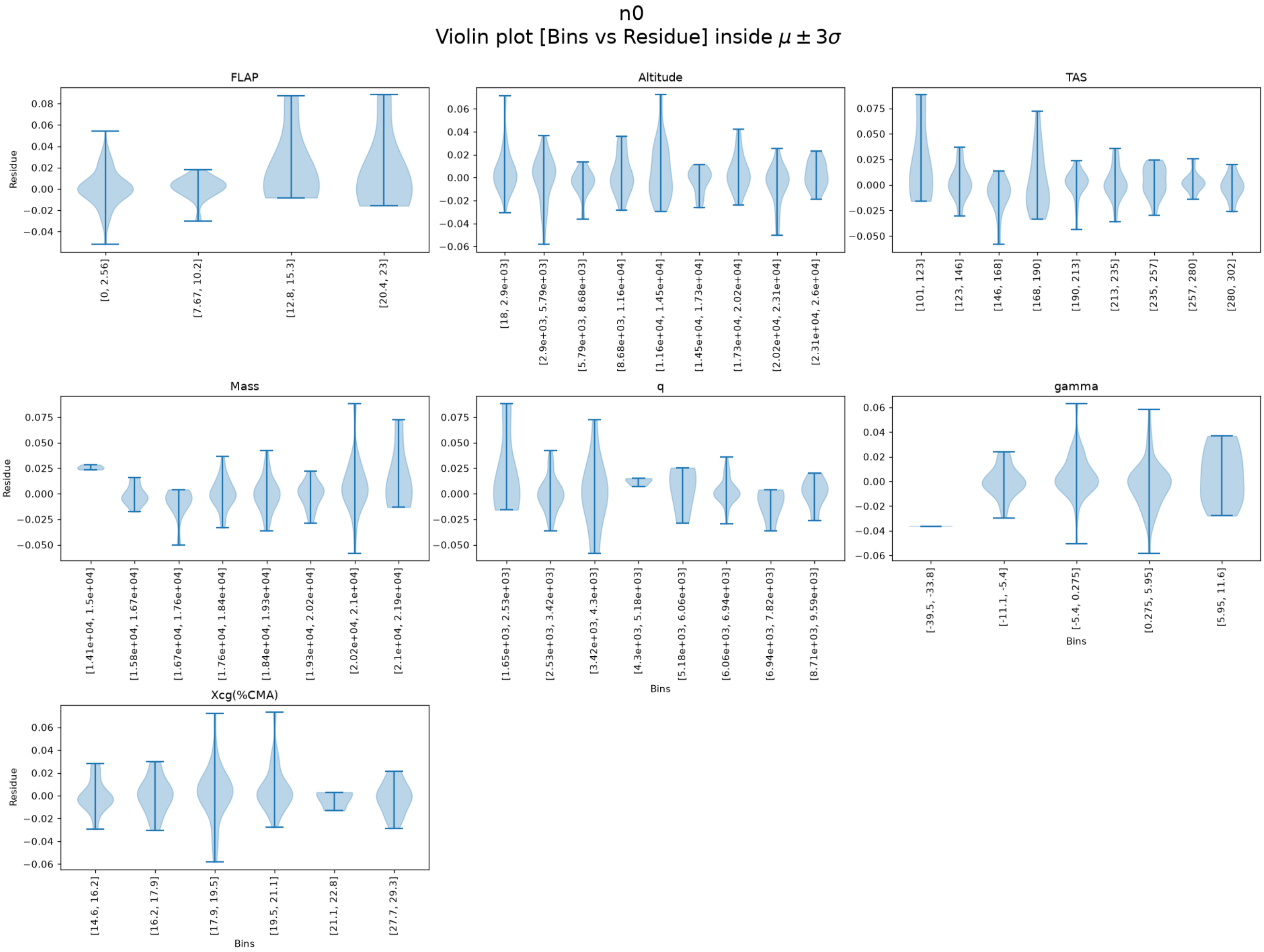
Residue — Turn



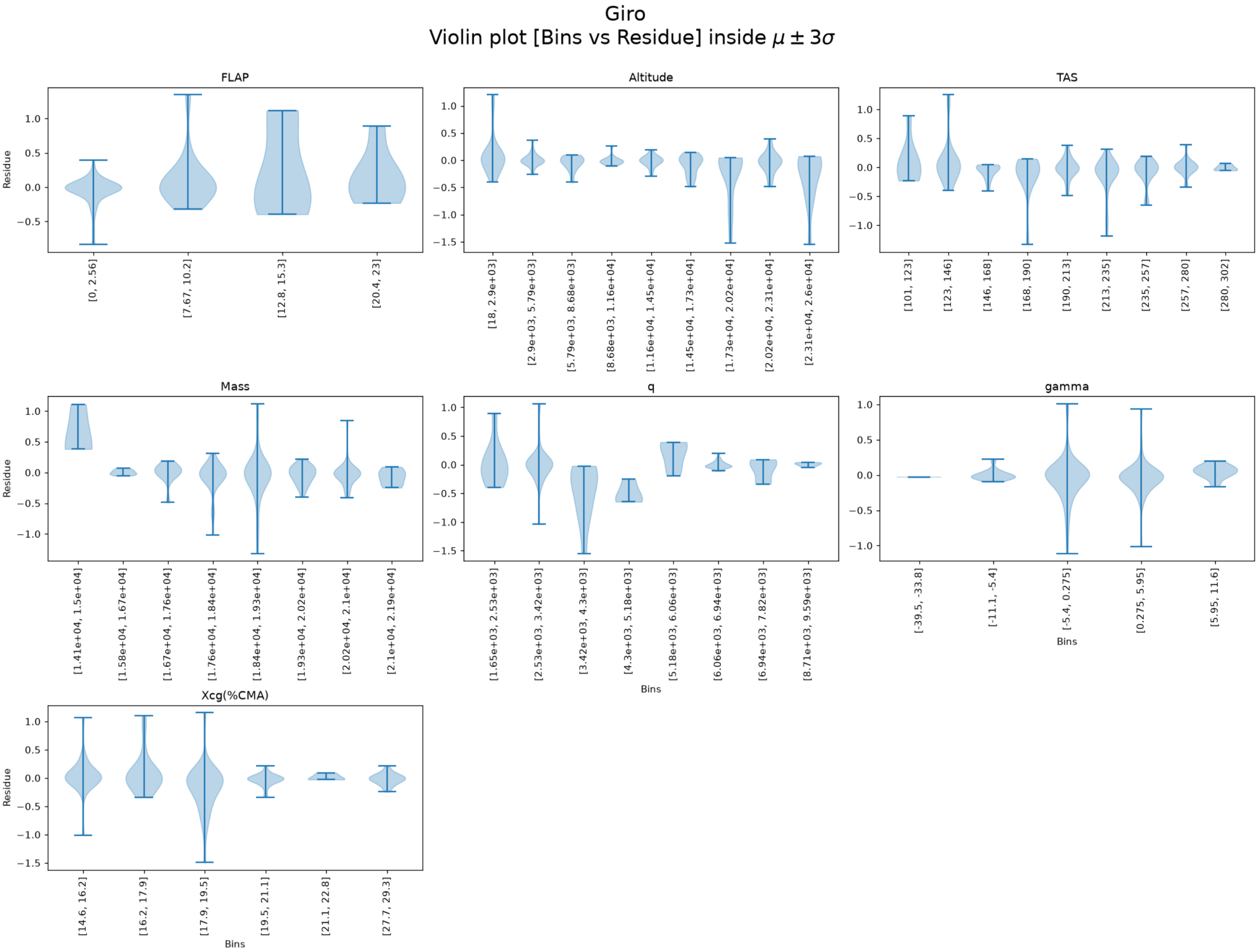
Residue — Frontal gust



Residue — n0

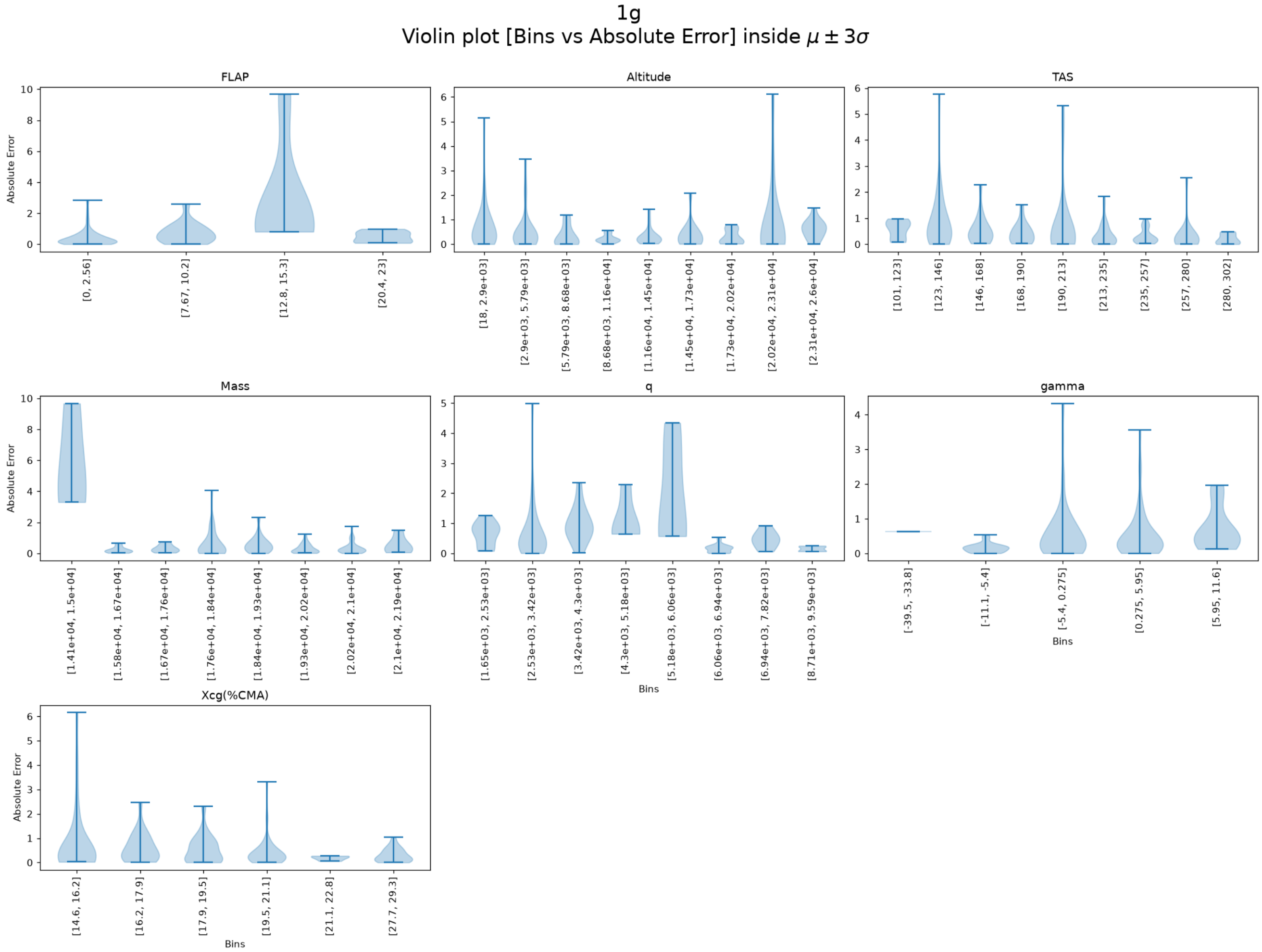

n0
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$
FLAP
Altitude
TAS
Mass
q
gamma
Xcg(%CMA)
Residue
Bins
[0, 2.56]
[7.67, 10.2]
[12.8, 15.3]
[20.4, 23]
[18, 2.9e+03]
[2.9e+03, 5.79e+03]
[5.79e+03, 8.68e+03]
[8.68e+03, 1.16e+04]
[1.16e+04, 1.45e+04]
[1.45e+04, 1.73e+04]
[1.73e+04, 2.02e+04]
[2.02e+04, 2.31e+04]
[2.31e+04, 2.6e+04]
[101, 123]
[123, 146]
[146, 168]
[168, 190]
[190, 213]
[213, 235]
[235, 257]
[257, 280]
[280, 302]
[1.41e+04, 1.5e+04]
[1.58e+04, 1.67e+04]
[1.67e+04, 1.76e+04]
[1.76e+04, 1.84e+04]
[1.84e+04, 1.93e+04]
[1.93e+04, 2.02e+04]
[2.02e+04, 2.1e+04]
[2.1e+04, 2.19e+04]
[1.65e+03, 2.53e+03]
[2.53e+03, 3.42e+03]
[3.42e+03, 4.3e+03]
[4.3e+03, 5.18e+03]
[5.18e+03, 6.06e+03]
[6.06e+03, 6.94e+03]
[6.94e+03, 7.82e+03]
[8.71e+03, 9.59e+03]
[-39.5, -33.8]
[-11.1, -5.4]
[-5.4, 0.275]
[0.275, 5.95]
[5.95, 11.6]
[14.6, 16.2]
[16.2, 17.9]
[17.9, 19.5]
[19.5, 21.1]
[21.1, 22.8]
[27.7, 29.3]

Residue — Giro

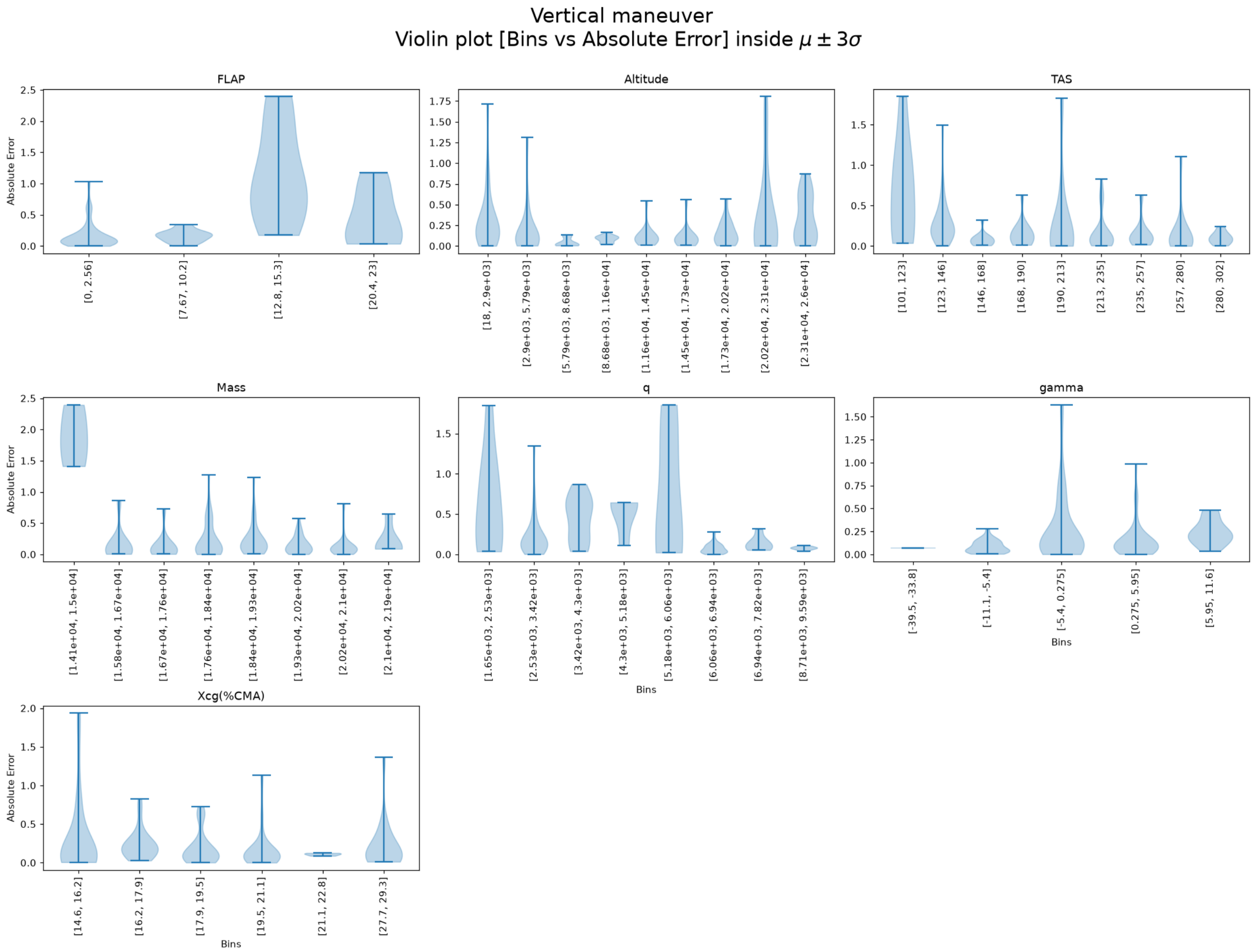


Absolute Error Violin Plots per Output (binned by input)

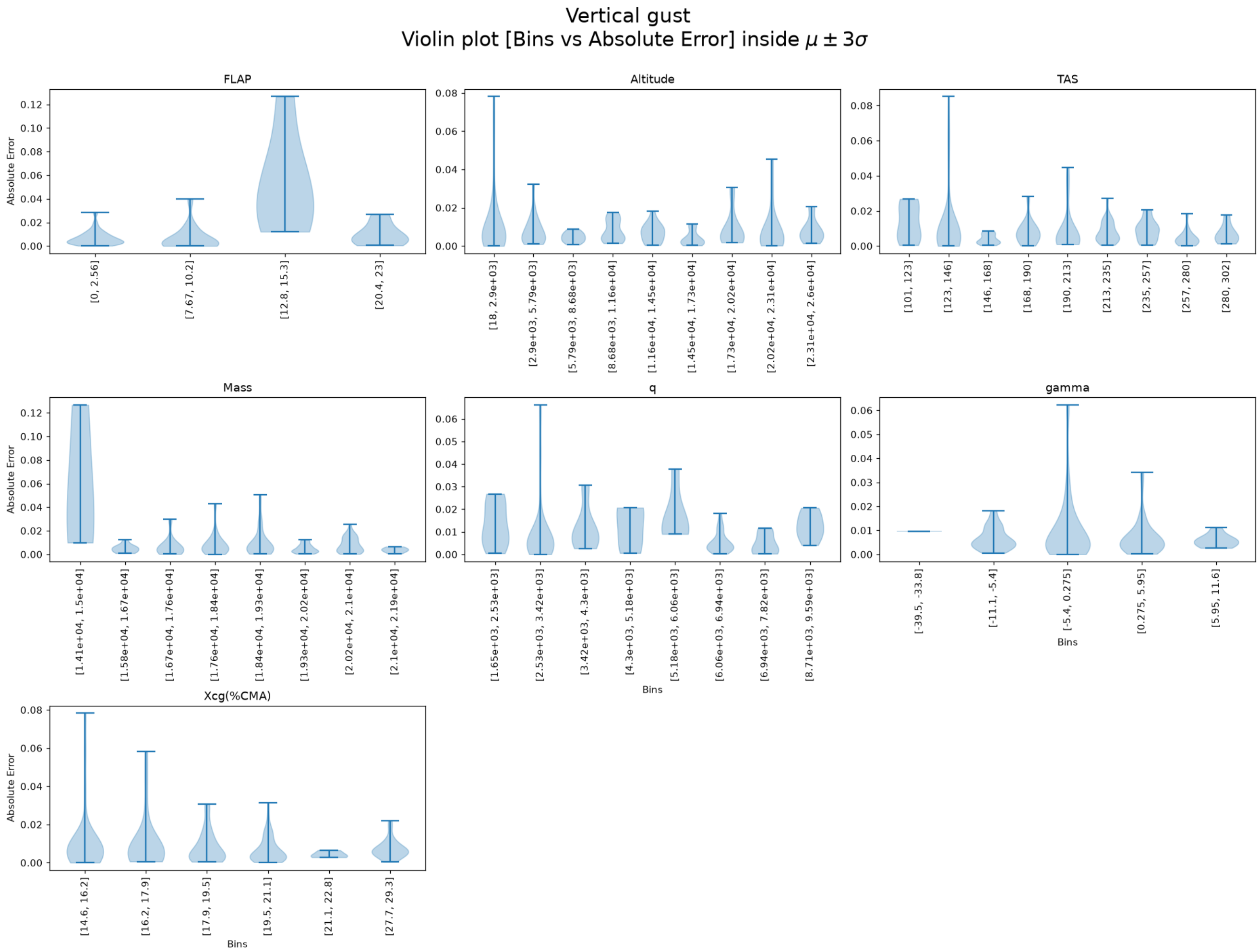
Absolute Error — 1g



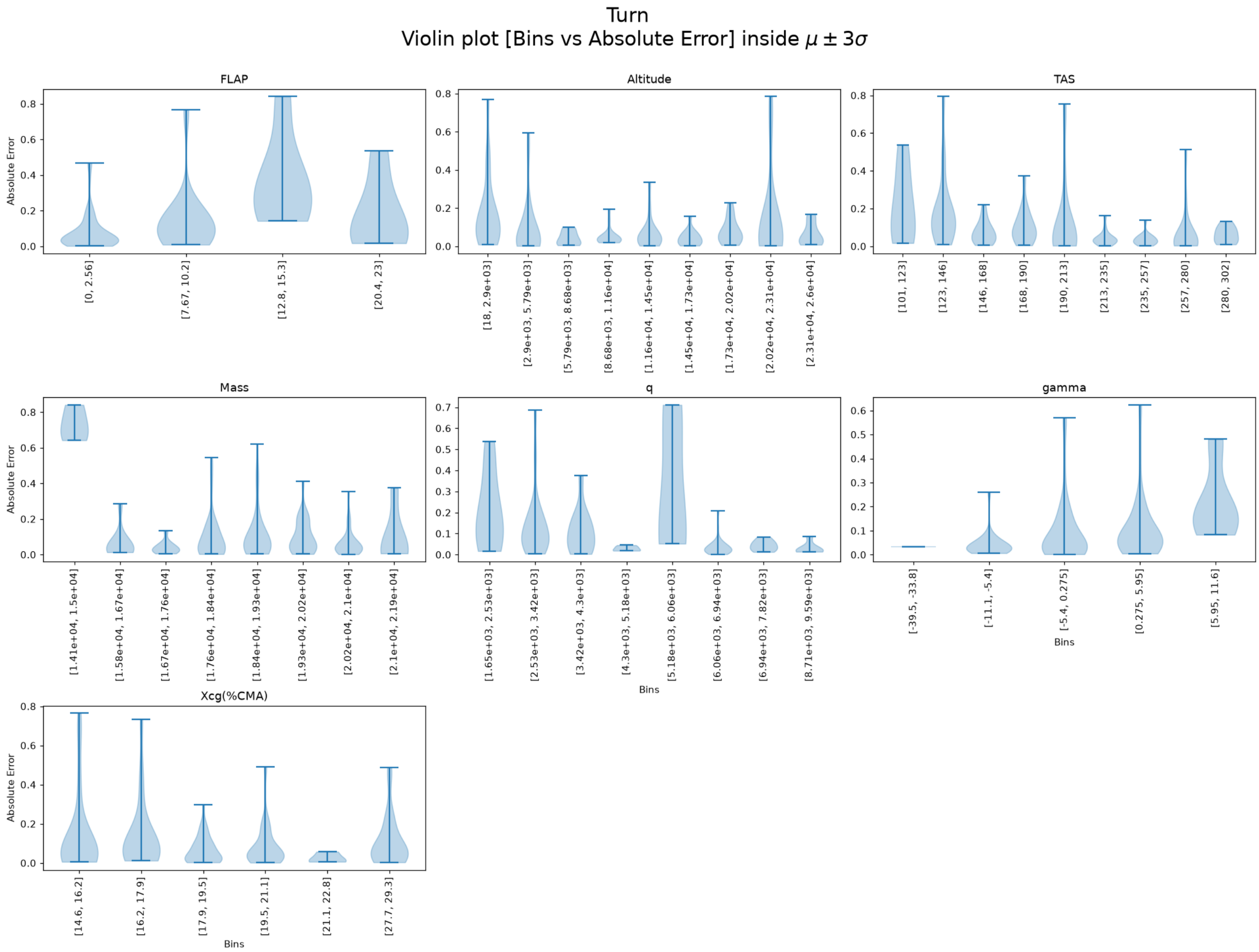
Absolute Error — Vertical maneuver



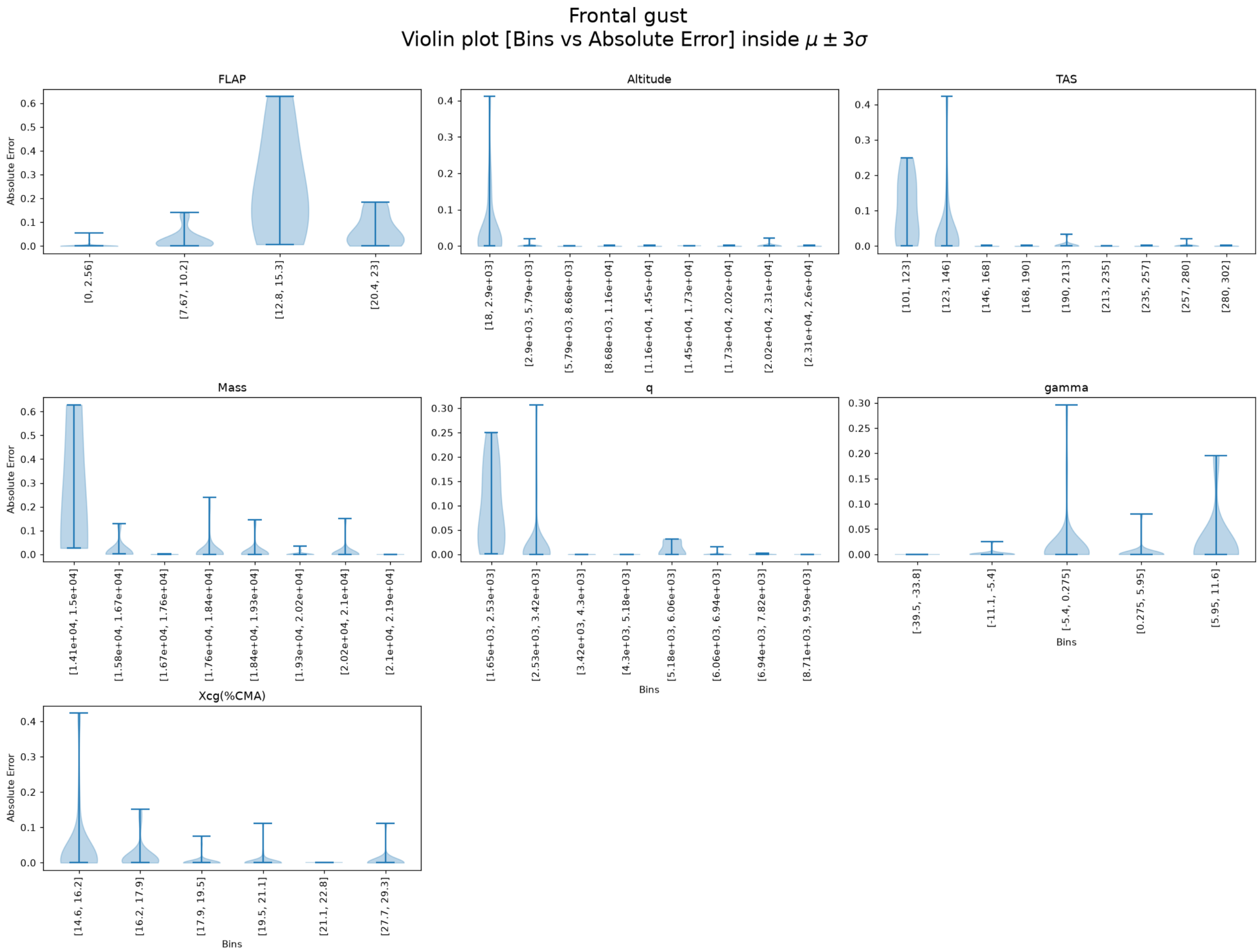
Absolute Error — Vertical gust



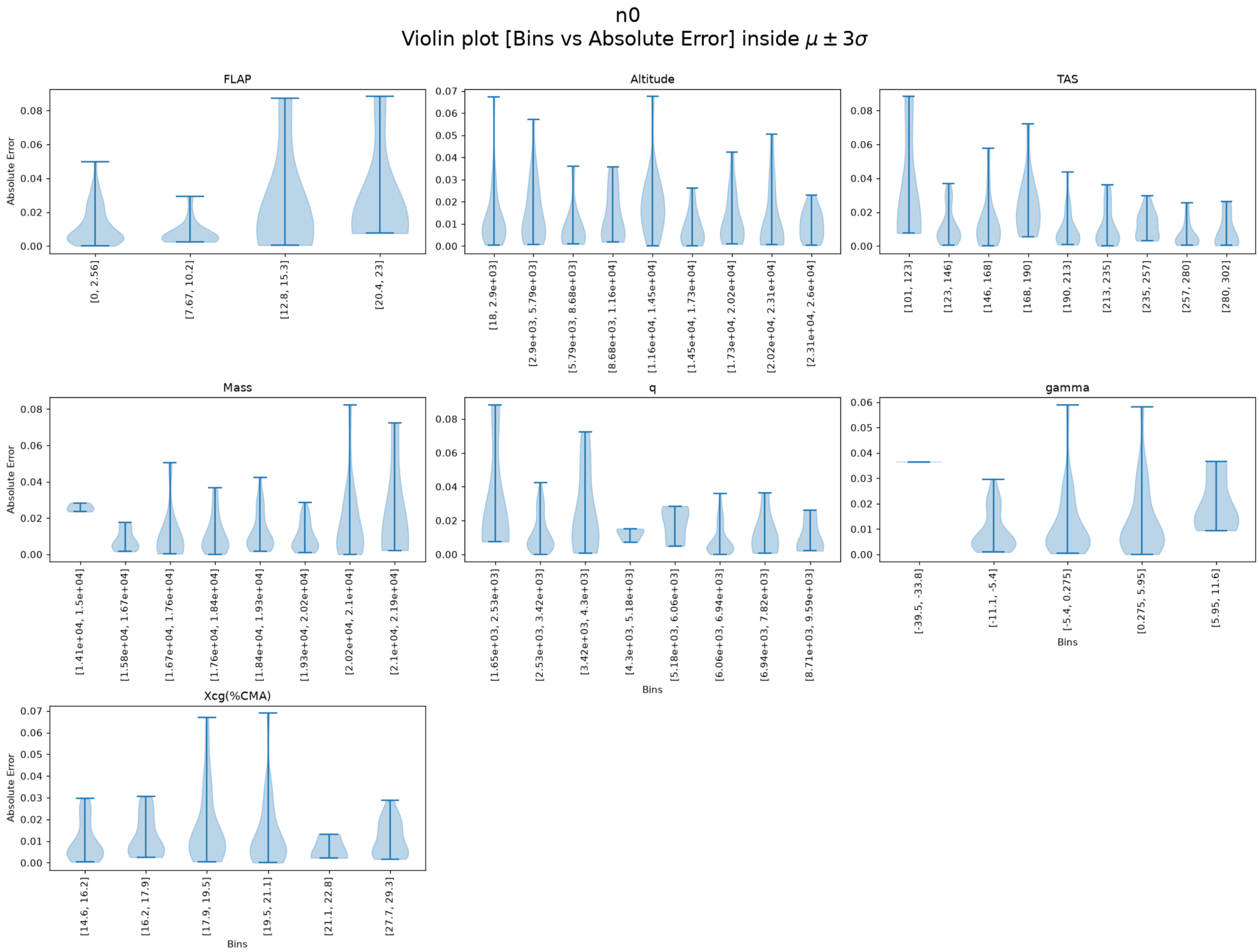
Absolute Error — Turn



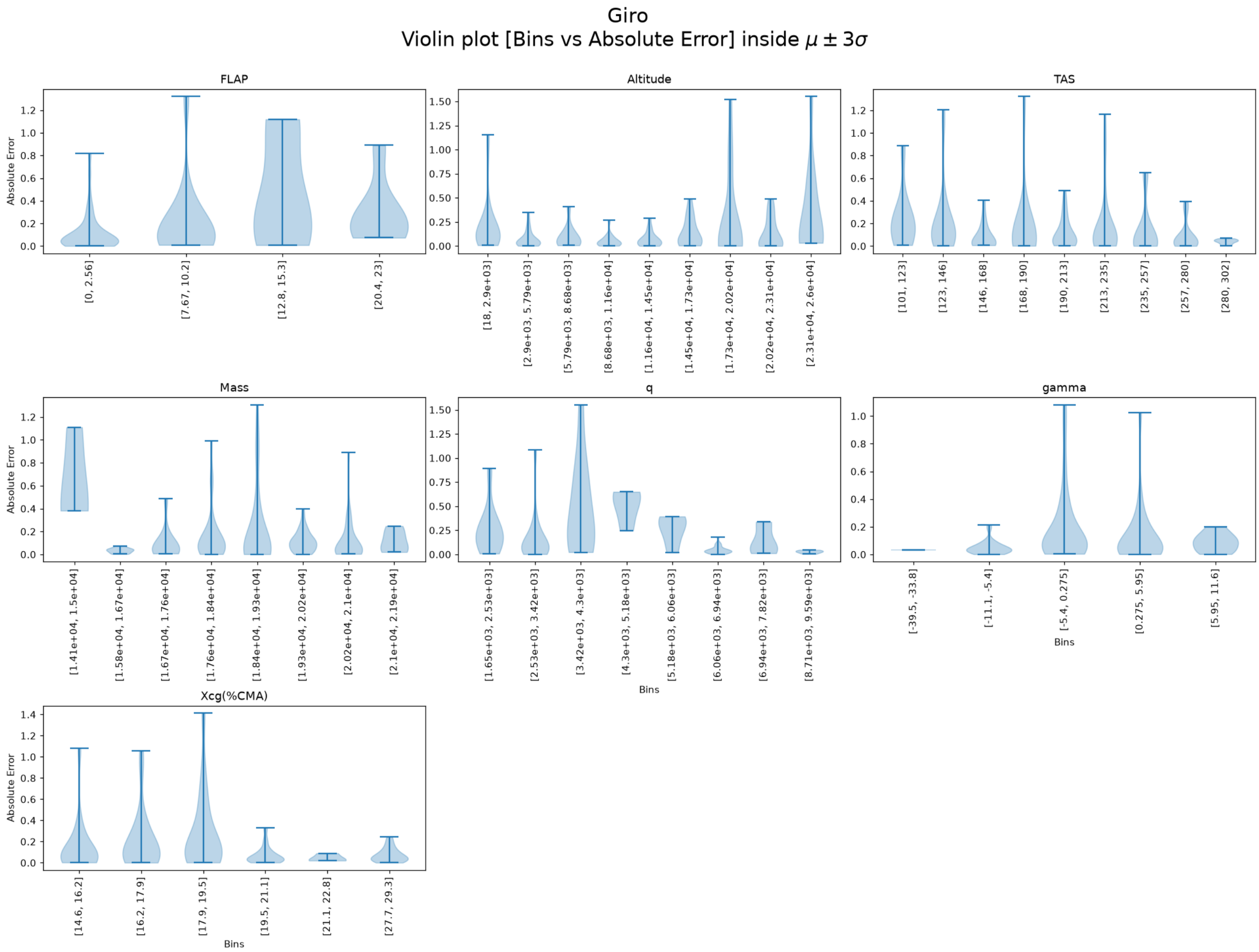
Absolute Error — Frontal gust



Absolute Error — n0



Absolute Error — Giro

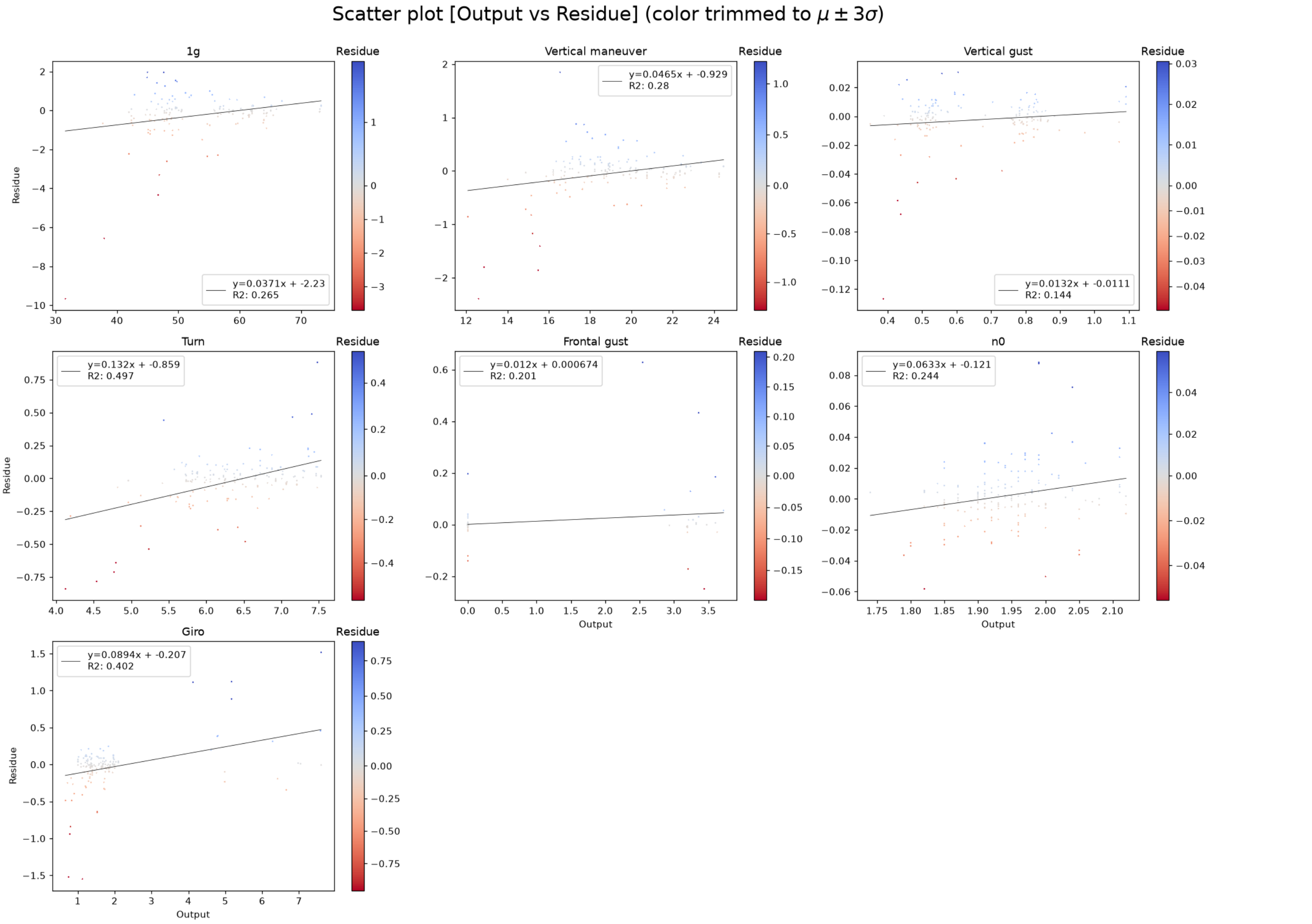


P(E|Y) — Error Conditional on Outputs

We study whether the error depends on the output magnitude. A significant trend (Pearson or Spearman $p < 0.05$) means the model is more accurate in some output ranges than others — a form of heteroscedasticity.

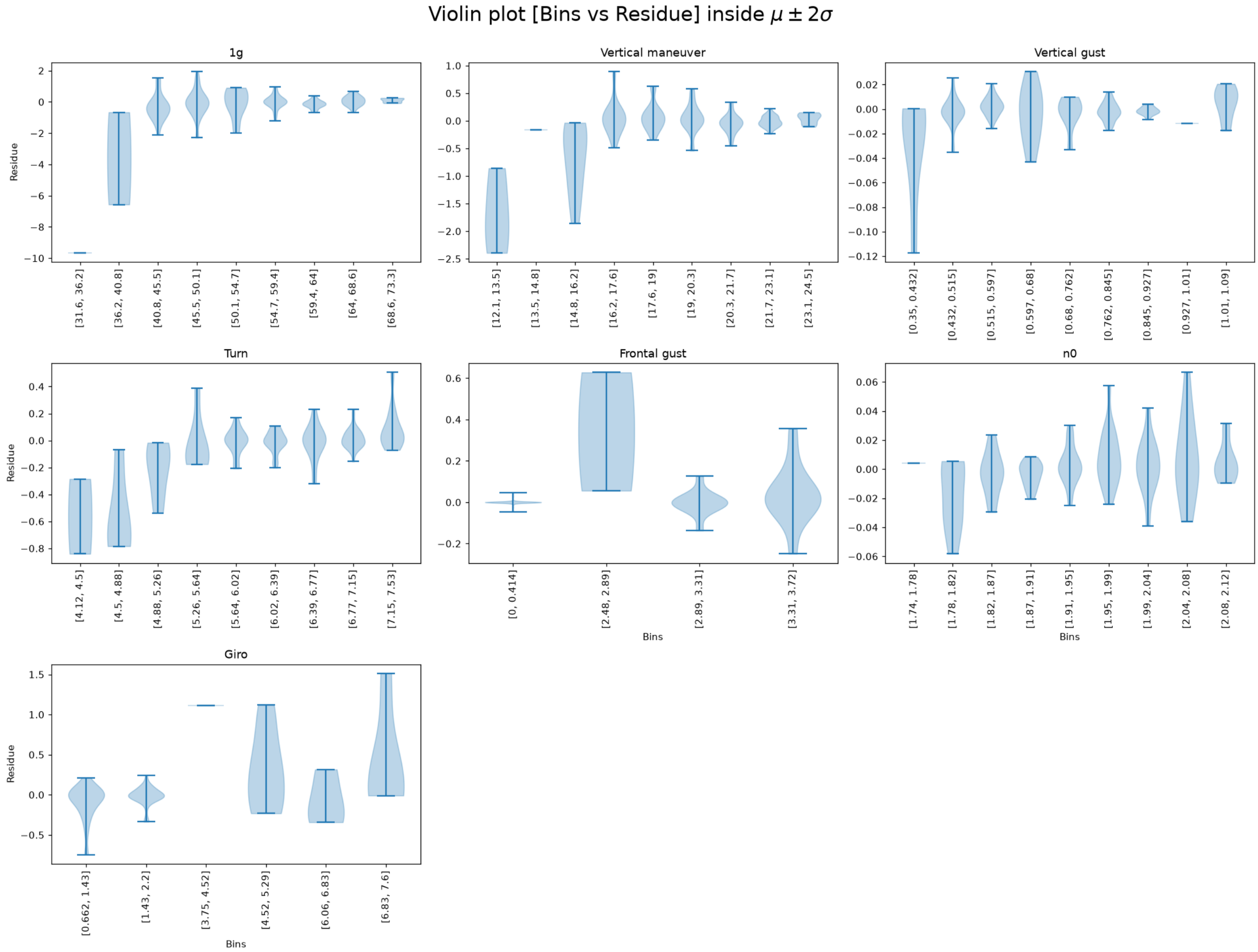
Residue vs True Output — Scatter

Pearson r and p-value shown in title. Red title = significant linear trend.



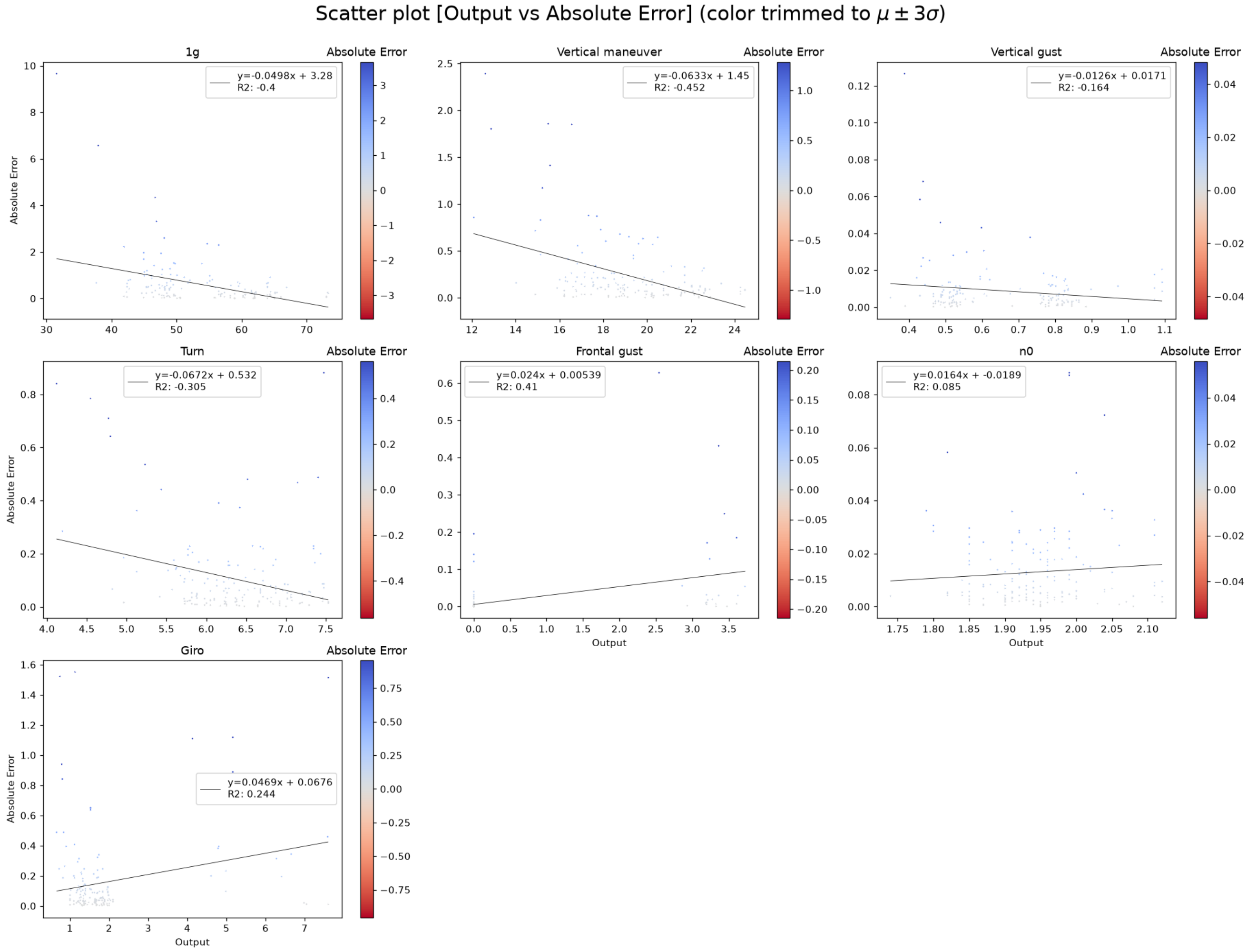
Residue Violin vs True Output Bins

Bins determined by Sturges rule. Median line dashed at 0.

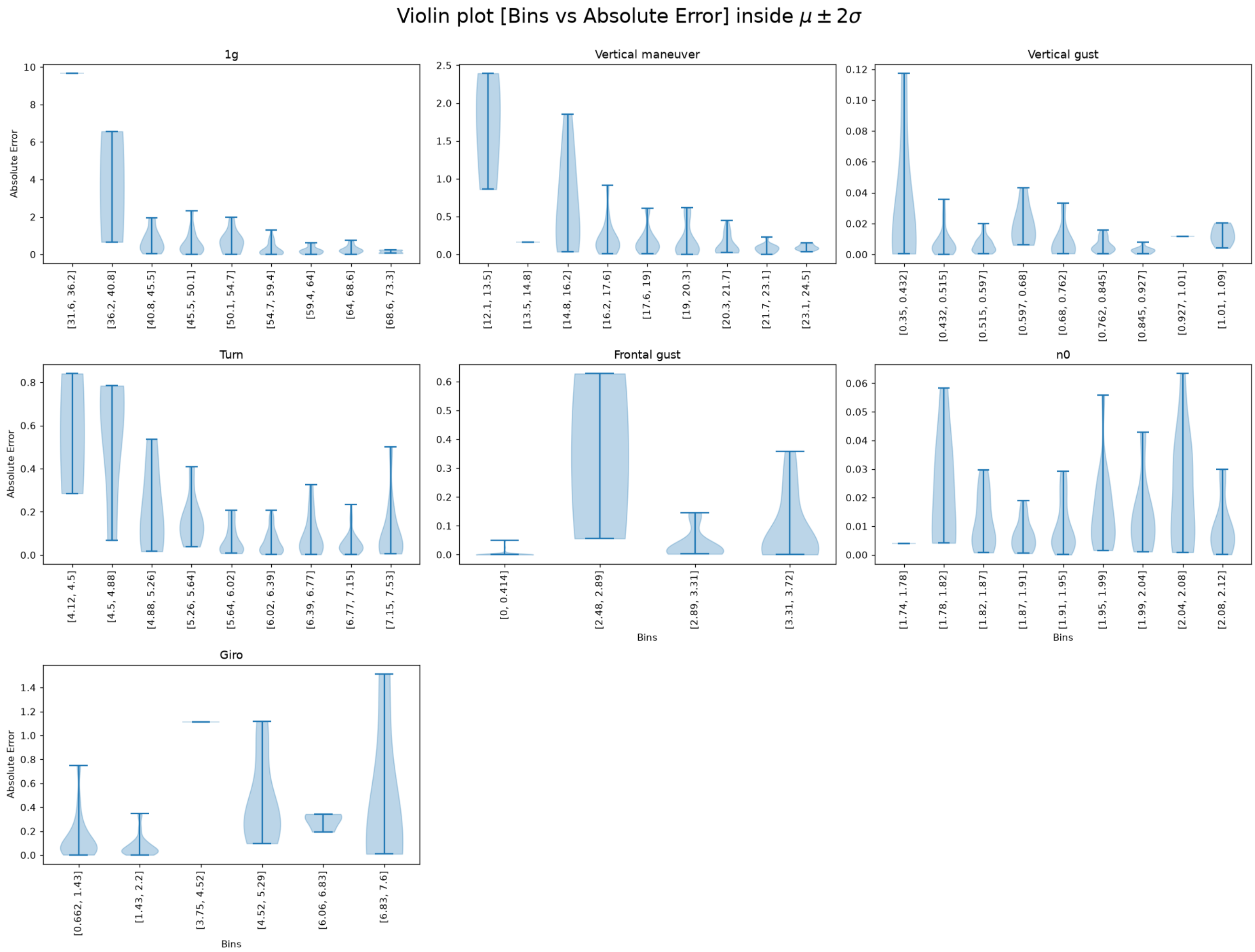


Absolute Error vs True Output — Scatter

Spearman r shown. Red title = significant monotonic trend.



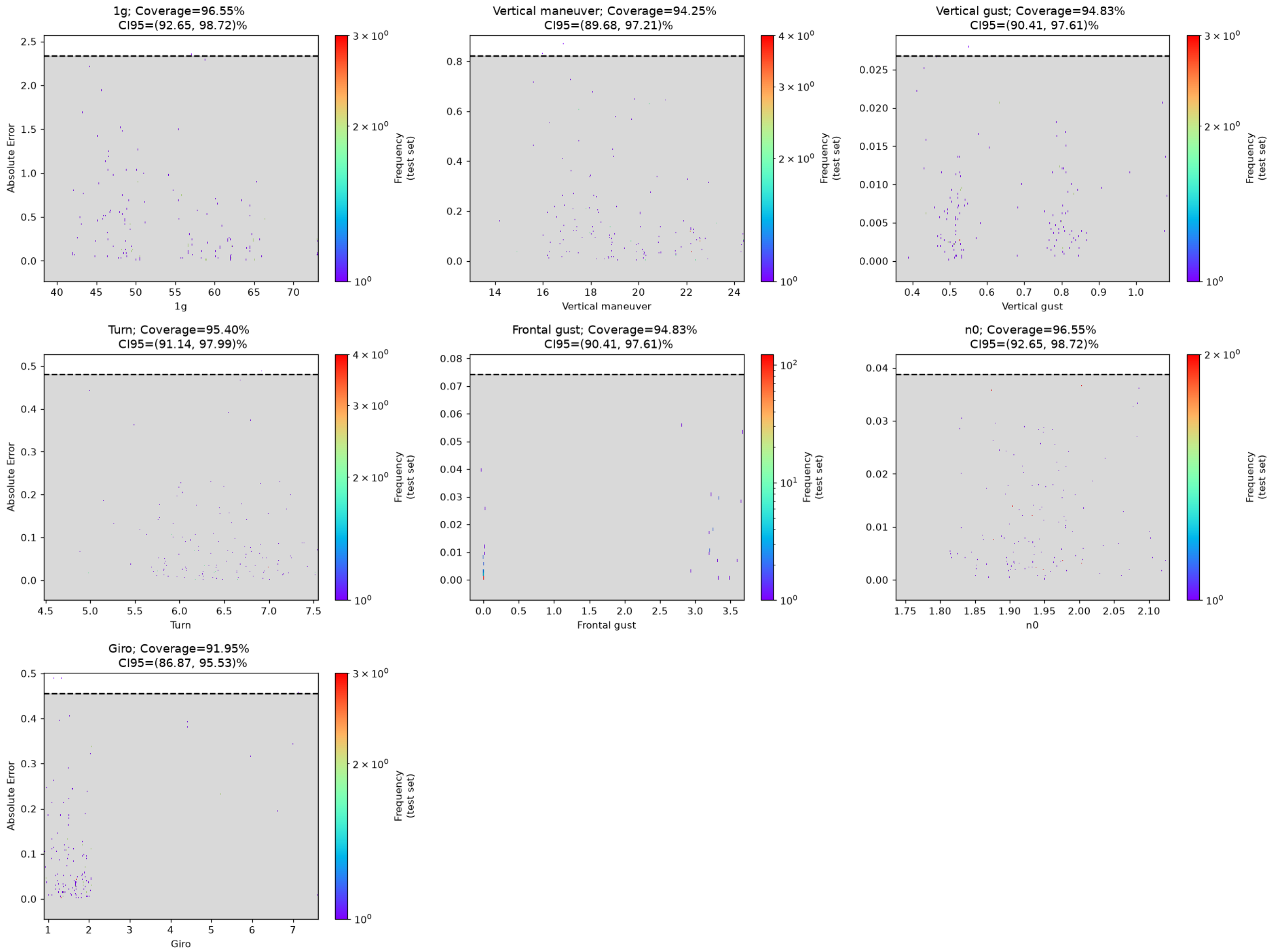
Absolute Error Violin vs True Output Bins



Uncertainty Analysis

A global binned uncertainty model (95th percentile, right-tailed) is trained on the validation split of the absolute error and evaluated on the held-out test split, as in the extended validation notebook. Coverage is the proportion of test points whose error falls inside the predicted bound; it should be close to 95%.

Coverage Plot



Coverage Table

	1g	Vertical maneuver	Vertical gust	Turn	Frontal gust	n0	Giro	TOTAL COV
Coverage	96.55	94.25	94.83	95.4	94.83	96.55	91.95	—
CI	[92.65, 98.72]	[89.68, 97.21]	[90.41, 97.61]	[91.14, 97.99]	[90.41, 97.61]	[92.65, 98.72]	[86.87, 95.53]	—

Empirical coverage of the 95th-percentile uncertainty bound on the test split.